

An Explainable AI Pipeline for Academic Integrity Verification

Transparent AI-Text Detection with Argument-Aware Question Generation

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Transparent AI-Text Detection with Argument-Aware Question Generation

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Abstract

Universities increasingly rely on automatic detectors to judge whether student work was written with generative AI. Most detectors return a single percentage with nothing behind it, which a lecturer cannot defend if a student challenges it, and which says nothing about whether the student understands the work they submitted. This dissertation develops an explainable pipeline for academic integrity verification. It detects likely AI-generated text, explains which features drove each decision, extracts the claims in a flagged submission, and generates verification questions tied to those claims so a lecturer can check understanding in a short conversation. The detector was trained on a balanced corpus of 640 human essays from the British Academic Written English corpus and 640 length-matched and topic-matched AI essays generated locally with Llama 3.1. An initial model reached a perfect score, which was treated as a warning rather than a result: a reproducible audit found the corpus leaked formatting markup, removed it from both classes, and confirmed that the classes remain separable on writing style alone (a function-words-only model still reaches 99.5 percent; the cleaned transformer scores an F1 of 0.99). The decisions are then explained, and the explanations tested: token-level attributions on the transformer prove diffuse and only weakly faithful, while SHAP over stylometric features passes an ablation-based faithfulness test and gives the lecturer-facing explanation. On an external benchmark the detector transfers to unseen generators (F1 0.97) but degrades across domains (F1 0.79), where its errors concentrate on falsely flagging human academic text, which quantifies the fairness risk that motivates the project. A first slice of the question-generation pipeline turns a flagged essay into claim-grounded verification questions with sentence-level provenance, and a judge-free discrimination simulation evaluates them, showing that questions forcing a student to reproduce their own reasoning discriminate better than questions answerable from general knowledge. A commercial-versus-local comparison over fourteen essays finds a small but statistically significant commercial advantage (paired difference 0.036, $p = 0.040$) with the local open model competitive before any fine-tuning, and a three-judge LLM-as-judge panel (Gemini, Claude, GPT) whose members neither agree with each other (Krippendorff's alpha -0.25) nor correlate positively with the objective measure, supporting the decision to anchor judges to the simulation rather than trust them. A QLoRA fine-tune of a small local model on an educational-question corpus appeared on the same simulation to

beat the commercial model, but inspection of its output showed it had collapsed into degenerate multiple-choice stems that game the metric: a negative result that redirects the local-model work toward open-ended training data and shows the discrimination measure needs a well-formedness guard. Re-fine-tuning on open-ended questions (SQuAD) confirmed the diagnosis, removing the degeneracy entirely and yielding a real but modest gain over the base model (0.10 vs 0.03, $p = 0.0003$) on well-formed questions, still short of a strong verification-question generator. This document is a working progress draft for supervisor review and will be revised further before final submission.

Acronyms

Acronym	Definition
AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
GenAI	Generative Artificial Intelligence
BAWE	British Academic Written English (corpus)
TTR	Type-Token Ratio
POS	Part of Speech
BIO	Begin-Inside-Outside (sequence labelling scheme)
QLoRA	Quantised Low-Rank Adaptation
SHAP	SHapley Additive exPlanations
IG	Integrated Gradients
FPR	False-Positive Rate
HPC	High-Performance Computing

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Chapter 1: Introduction

1.1 Overview

This dissertation builds an explainable pipeline that helps a lecturer check the integrity of a student submission without having to treat a detector's score as the final word. It does two connected things. First it decides whether a piece of writing is likely AI-generated, and rather than returning a bare percentage it shows which features of the writing drove that decision. Second, when a submission is flagged, it reads the student's own argument, pulls out the specific claims they made, and turns those claims into verification questions a lecturer can ask in a short conversation. The aim is not to catch students out. It is to give the lecturer transparent, defensible evidence, tied to the student's own words, that a fair decision can rest on.

Two design choices run through the whole system. Every stage is meant to be explainable, so a lecturer can see and justify why a submission was flagged and where each question came from, instead of pointing at a number they cannot account for. And the question generation is built twice, once with a commercial large language model and once with a smaller open model that runs on a single laptop, so the project can ask a question that matters to any institution: how close can a locally run, low-cost model get to an expensive commercial one for this task (Oketch et al., 2025). That comparison is a central contribution.

The intended user is a lecturer on a large module, somewhere between 150 and 300 students, who has neither the time to prepare a fair set of verification questions for every flagged essay nor a good way to defend a black-box score if a student challenges it. The system is aimed squarely at that person. It is designed to save time, to treat students fairly, including students who do not write in English as a first language and whom existing detectors tend to penalise (Liang et al., 2023), and to produce something a lecturer can stand over.

1.2 Background and context

Over the last few years generative AI tools have moved from a curiosity to something most students have used at least once (Jin et al., 2024). Drafting an opening paragraph, rephrasing an awkward sentence, or getting a hard idea explained is now a few seconds of typing into a chatbot. That is genuinely useful for learning, but it has also put pressure on a basic academic question: is this piece of work the student's own, and does it reflect their own understanding?

Universities responded quickly, and a market of AI-text detectors appeared almost as fast as the writing tools did (Wu et al., 2025). Broadly these detectors fall into two families. Some are zero-shot or statistical: they use a language model's own probabilities to judge how predictable a passage is, on the assumption that machine text sits in smoother, more probable regions than human text (Mitchell et al., 2023). Others are supervised: a transformer is fine-tuned on labelled human and AI examples and then classifies new text (He et al., 2021). Commercial services usually combine such methods behind an interface and return a single figure, an AI-likelihood percentage.

The trouble is not that these tools never work. It is that they were built fast, they are opaque, and they are fragile in the situations that matter most. Their accuracy drops sharply on text they were not trained for, and they can be evaded by simple paraphrasing (Krishna et al., 2023). More seriously for a university, they hand back a number with nothing behind it, and there is growing evidence that the errors are not spread evenly: writing by non-native English speakers is flagged as AI far more often than it should be (Liang et al., 2023). A tool that is confident, unexplained, and unfair is a poor basis for an academic-integrity decision.

1.3 Problem statement

The core problem I keep returning to is that current detectors give a lecturer a percentage and stop there. Someone who sees "82 percent AI" has no way to explain that number if the student pushes back, and no real way to know whether it is even right for this particular essay. That is a weak and risky position to accuse a person from, and the cost of being wrong falls on the student, whose record and standing are at stake. A score with no reasons behind it cannot carry that weight.

There is a fairness problem on top of the opacity. Because the errors are biased, the students most likely to be wrongly flagged are often those who are already disadvantaged, including people writing in a second language (Liang et al., 2023). A system that quietly repeats that bias, and cannot even show its reasoning, is worse than no system at all.

The deeper problem is that even a correct flag does not answer the question a lecturer actually cares about. Knowing that text is probably AI-generated does not tell you whether the student understands the material they handed in. One student might have used a tool heavily and still grasp the argument; another might have written every word themselves and

understood little. A flag is a suspicion about how a text was produced, not evidence about what the student knows. What a lecturer needs, and what no detector provides, is a fair way to move from that suspicion to a check of understanding.

Closing that distance is the problem this project takes on: make the detection transparent and defensible, and then turn a flag into specific, source-grounded questions that test whether the student can account for their own work.

1.4 Motivation

I am aiming this at the lecturer who runs a large module and has no time to sit with each flagged essay and work out a fair way to check it. That person needs something they can defend, that treats students fairly, and that saves them time rather than adding to the pile. The fairness side matters to me in particular, because the research already shows these detectors can be biased against people who do not write in English as a first language (Liang et al., 2023), and I do not want to build something that quietly repeats that. A system that is transparent about its reasoning is also a system whose mistakes can be seen and argued with, which is exactly what a fair process needs.

1.5 The verification gap

So there is a gap. On one side is detection, which is improving but stays a black box. On the other side is the question that actually matters: does this student understand their own submission? Nothing I found connects the two. Nothing takes a flag and turns it into specific questions, drawn from the student's own claims, that a lecturer can ask in a short conversation to see whether the understanding is there. That gap is what this project tries to fill.

1.6 Research question and objectives

The main question I am asking is this: how can an explainable AI pipeline be designed to support lecturers in academic integrity verification, by combining transparent AI text detection with argument-aware question generation, and how far can locally fine-tuned open-source models match commercial ones at producing interpretable, defensible outputs for this task.

To get there the objectives are roughly:

- detect AI-written text in a way that can be explained, not just scored;
- show which words and features drove each decision, and test that those explanations are faithful;
- pull out the claims and evidence in a flagged essay and tie each one to its source;
- generate verification questions from those claims, and compare a commercial model against a local open-source one;
- label each question by cognitive level as a quality check; and
- evaluate the whole thing with objective measures rather than opinion.

1.7 Scope and boundaries

To keep this doable in the time I have, the scope is deliberately tight. The detector works with two classes only, human and AI. I dropped the earlier idea of partial-AI classes because they overlap too much to separate reliably; that may come back later as future work if the two-class version works well. The pipeline has six parts (detection, explainability, argument mining, question generation, a Bloom's level check, and the output guide), and the datasets are fixed in advance. There are no human participants in the study, which keeps it clear of needing ethics approval and keeps the focus on the system itself.

1.8 Contributions

What I think this project adds:

- a pipeline that goes from a detection flag to transparent, defensible evidence, instead of stopping at a score;
- verification questions that are tied back to the student's own writing, so a lecturer can see where each one came from;
- a direct comparison of a locally run open model against a commercial one for this task, which matters for any institution thinking about cost and control;
- an objective way to measure whether a question really tests understanding; and
- an honest look at how the detector treats native versus non-native writers.

1.9 Dissertation outline

The rest of the document is laid out as follows.

Chapter 2 reviews the recent literature the project draws on: AI-text detection and its benchmarks, stylometric features, explainability and faithfulness, argument mining, question generation and its evaluation, and the fairness evidence that motivates the design.

Chapter 3 sets out the detection methodology and the first results: the dataset, how the matched AI essays were built, and the audit that found and removed a corpus artefact behind an initially perfect score.

Chapter 4 describes the implementation of the components built so far, with the engineering decisions that the 8 GB laptop budget forced.

Chapter 5 explains the detector's decisions and tests whether those explanations are faithful, comparing token-level attributions on the transformer against SHAP over the stylometric features.

Chapter 6 tests robustness: how the detector transfers to generators it never saw and to other kinds of text, and what its failures mean for fairness.

Chapter 7 presents the core contribution: turning a flag into verification questions drawn from the student's own claims, with sentence-level provenance and a Bloom's level on each question. It builds the first thin slice, then replaces its stand-ins with the trained Bloom's classifier and the trained claim extractor, assembles the lecturer's Verification Interview Guide, and fine-tunes the local backend.

Chapter 8 evaluates the questions. It builds the judge-free discrimination simulation, runs the commercial-versus-local comparison under two designs, validates a three-judge LLM panel against the objective measure, and reports the fine-tuning data-format experiment together with the quality audit that caught a metric-gaming failure.

Chapter 9 discusses what the results mean: the answer to the research question, the methodological lesson the failures taught, the case for verification over accusation, the limitations, and the implications for practice. Chapter 10 concludes and sets out future work, and the references follow.

Chapter 2: Literature Review

2.1 Scope of the review

This review covers the parts of the literature the project draws on and argues against: detecting AI-generated text, the stylometric features behind the detector, explainability and how to test whether an explanation is honest, argument mining, question generation, how question quality is judged, and fairness. The sections follow the pipeline so the review builds toward the gap the project fills. I anchored the search on the specific methods and datasets already chosen (the DeBERTa detector, the M4 benchmark, the Persuasive Essays corpus, Bloom's taxonomy, the answerability idea behind the evaluation) and then branched out to alternatives and criticisms.

2.2 Detecting AI-generated text

Work on detecting machine-generated text splits into two broad families, and a recent survey by Wu et al. (2025) organises the field along these lines, adding watermarking and human-assisted methods.

The first family is zero-shot and statistical. These methods use a language model's own probabilities rather than a trained classifier. The clearest example is DetectGPT (Mitchell et al., 2023), which observes that machine-generated passages tend to occupy negative-curvature regions of a model's log-probability surface, and turns that observation into a curvature test that needs no labelled data. The appeal is that nothing has to be trained; the weakness is that the signal is subtle and can be disturbed.

The second family is supervised. A pretrained transformer is fine-tuned on labelled human and AI examples and then classifies new text. The transformer this project uses, DeBERTa (He et al., 2021), improves on earlier encoders with a disentangled attention mechanism and was chosen as the primary detector for that reason, with RoBERTa (Liu et al., 2019) as a comparison. Supervised detectors are accurate in-domain but depend heavily on what they were trained on.

Progress is measured on shared benchmarks. The M4 corpus (Wang et al., 2023) collects human and machine text across several generators, domains, and languages, and the SemEval-2024 Task 8 shared task built on it (Wang et al., 2024). I use that benchmark to test

whether a detector trained on one generator and one kind of text transfers to unseen generators and unseen domains.

Two weaknesses matter for this project. First, detectors are brittle: Krishna et al. (2023) show that a strong paraphraser drops DetectGPT's accuracy from 70.3% to 4.6% without changing the meaning, and accuracy also falls under domain shift. Second, and more important here, commercial detectors return a single opaque score. A number with no reasons behind it is a weak basis for an academic-integrity decision, and that black-box problem is the starting point for this project.

2.3 Stylometric features

Alongside the transformer, the detector uses stylometric features, which describe how a text is written rather than what it is about: perplexity or predictability under a language model, burstiness (variation in sentence length and structure), type-token ratio, sentence-length variance, and part-of-speech distributions. Each is meant to capture a way in which fluent machine text tends to differ from human writing, being smoother and less varied.

There is direct evidence that these surface properties carry a real, and sometimes harmful, signal. Liang et al. (2023) attribute the misclassification of non-native English writing to its lower perplexity, that is, its more limited linguistic variability, which makes it look more machine-like to a perplexity-based detector. That finding is a double lesson: stylometric features do separate the classes, but they can also encode bias, so the pipeline treats them as interpretable evidence whose behaviour must itself be checked. Recent work confirms that hand-crafted features (perplexity, readability, error-based and lexical features) can classify AI-generated and even AI-rephrased text, alone and alongside neural models (Mindner et al., 2023), which supports the hybrid design used here.

2.4 Explainability and faithfulness

The explainability layer is the point of the project, so this section covers both the attribution methods and, just as important, how to check whether an explanation is honest.

Two attribution methods are used. Integrated Gradients (Sundararajan et al., 2017) attributes a prediction to its input features while satisfying two axioms, Sensitivity and Implementation Invariance, and is applied here to the transformer's tokens. SHAP (Lundberg and Lee, 2017)

assigns each feature a Shapley-value importance and is applied to the stylometric features, where it gives a lecturer-facing, feature-level explanation. Attention weights are sometimes offered as explanations too, but that use has been criticised (Jain and Wallace, 2019), which is one reason this project does not rely on attention.

An explanation is only useful if it is faithful, meaning the things it highlights are actually what drove the decision. DeYoung et al. (2020) introduced the ERASER benchmark and the comprehensiveness and sufficiency metrics, which measure faithfulness by removing or keeping the features an explanation calls important and seeing whether the prediction moves. This project uses exactly that kind of ablation test, and it is what showed the transformer's token-level attributions to be diffuse and only weakly faithful, while the stylometric SHAP explanation held up.

2.5 Argument mining

To turn a flag into questions grounded in the student's own argument, the pipeline has to extract the claims and evidence in an essay. The foundational resource is the Persuasive Essays corpus of Stab and Gurevych (2017), which annotates argument components (claims, premises) and the relations between them, and frames the task as token-level sequence labelling plus relation classification. The argument-mining component here follows that setup; more recent work applies transformer ensembles, and even LLM refinement, to argument component classification across several corpora (Pietron et al., 2024).

Provenance is the reason argument mining matters here rather than mere extraction. Each extracted claim must point back to the exact source passage it came from, so that a generated question can be tied to where it originated and a lecturer can defend it. Extraction without provenance would reintroduce the black-box problem the project is trying to remove.

2.6 Automatic question generation

Question generation has moved from neural models that generate a question from a passage to large language models prompted to do the same, with growing interest in controllable and grounded generation (Guo et al., 2024). For verification, grounding is the crucial property: a question is only defensible if it can be traced to a specific claim in the student's own text, so the generator works per extracted claim and records the source sentences.

The comparison at the centre of this project, a commercial model against a locally run open model, sits in the wider question of cost and control in LLM deployment. In the closest study to this setting, Oketch et al. (2025) compared closed and open LLMs for automated essay scoring and found open models such as Llama 3 and Qwen2.5 comparable to GPT-4 in performance and fairness at up to 37 times lower cost. An institution that could run verification on a laptop rather than a paid API would have a cheaper and more controllable option, if the quality holds, which is the empirical question the project tests.

2.7 Bloom's taxonomy and cognitive level

Questions are labelled by cognitive level as a quality control, using the revised Bloom's taxonomy of Anderson and Krathwohl (2001), which orders cognitive processes from Remember and Understand through Apply and Analyse to Evaluate and Create. The taxonomy gives a principled way to describe what a question demands of a student.

For the educational side there is a directly relevant dataset. EduQG (Hadifar et al., 2022) provides 3,397 questions with their source documents, of which 903 are annotated with a Bloom's cognitive level, which makes it suitable for training and checking the Bloom's classifier used here. Automatic classification of question cognitive level is an active research task, where approaches from classical classifiers through transformers to large language models have been applied (Kumar et al., 2025).

2.8 Evaluating generated questions

The project's main evaluation asks whether a question genuinely tests understanding, and the closest prior idea is answerability-based, reference-free evaluation. RQUGE (Mohammadshahi et al., 2023) scores a generated question not by comparing it to a reference question but by whether a question-answering module can answer it from the given context. The discrimination simulation used in this project is in the same spirit: it asks whether a model can answer a question with the source versus without it, and treats the gap as the question's discriminative value.

The supplementary evaluation uses an LLM-as-judge, which is increasingly common but has known reliability and bias problems, including position, verbosity, and self-enhancement biases, so its output has to be checked against something independent before it counts

(Zheng et al., 2023). Cross-model agreement is quantified with Krippendorff's alpha (Hayes and Krippendorff, 2007), and anchored to the objective discrimination results.

2.9 Fairness and bias in detection

Fairness is not a side issue for this project; it is one of its motivations. Liang et al. (2023) tested seven widely used GPT detectors on TOEFL essays by non-native English writers and found an average false-positive rate of 61.3%, against about 5.1% for essays by native writers. In other words, the detectors systematically misclassified competent non-native writing as AI-generated. Chapter 6 measures the same failure mode directly on my own detector and treats it as evidence, not a footnote, which is why transparency and defensibility run through the design.

2.10 Summary and the gap

Detection is improving but stays opaque and, on the evidence above, sometimes unfair. Explainability methods exist and can be tested for faithfulness. Argument mining can recover a student's claims with provenance. Question generation and its evaluation are maturing, and Bloom's taxonomy gives a way to grade cognitive level. What is missing is anything that joins these up: nothing takes a detection flag and turns it into transparent, source-grounded verification questions that let a lecturer fairly check whether a student understands their own work. That gap, and the practical question of whether a local model can do the job as well as a commercial one, is what this project addresses.

Chapter 3: AI text detection (methodology and first results)

3.1 What this component has to do

The first stage of the pipeline decides whether a piece of submitted writing is the student's own or was produced by a generative model. Everything downstream depends on it, but the score on its own is not the point of the project. The point is a decision a lecturer can defend. So I am as interested in why the model decides what it decides as in how often it is right.

3.2 The detection corpus

I needed text that is labelled human or AI and that does not give the answer away for the wrong reason. The human side is a stratified sample of 640 real student essays from BAWE (Alsop and Nesi, 2009; the sampling is described in the sample design note). For the AI side I generated one essay per human essay with Llama 3.1 8B (Grattafiori et al., 2024) running locally, matched on topic, keywords and target length. Length matching matters: if the AI essays were systematically longer or shorter, a detector could score well just by counting words and would learn nothing about style. After generation the lengths line up closely (Pearson r about 0.98 between each AI essay and its human source, mean length ratio about 1.05), so length is not the signal.

That gives 1,280 essays, 640 human and 640 AI. I split them at the level of the student, not the essay, so no student appears in both training and test. If the same person's writing sat on both sides of the split, the model could memorise that person rather than learn the general difference. The split is 438 / 102 / 100 pairs for train / validation / test.

3.3 The detector

The detector is a fine-tuned transformer. DeBERTa-v3-base is the primary model and RoBERTa-base is a comparison, both fine-tuned to classify human (0) against AI (1) with the same settings and the same student-level splits. I report accuracy, precision, recall, F1 and the confusion matrix, plus a separate false-positive rate for native and non-native English writers, because wrongly flagging a non-native student as AI is the failure mode I most want to avoid. The second half of the hybrid, a stylometric feature model (perplexity, burstiness, vocabulary richness, part-of-speech mix), is built in Chapter 5 and the two halves are fused in Section 6.7; this chapter is the transformer half on its own.

3.4 The result, and why I did not trust it at first

The first trained detector scored a perfect F1 of 1.00 on the held-out test set, for both DeBERTa and RoBERTa. A perfect score is not a result to celebrate, it is a result to investigate. In a project that is about making detection defensible, presenting a number I could not explain would be the opposite of the point. So before using it I ran an audit (`src/detection/audit_detector.py`) to find out whether the score was real or a shortcut.

3.5 The audit

The audit asked four questions.

Is the test set contaminated? No. I checked that no student sits in more than one split, that each human and AI pair stays in the same split, and that no exact-duplicate text crosses from training into test. All three came back clean across 394 students. So the perfect score is not the model having seen the answers.

Is there a trivial giveaway in the text? Yes, and this is the important finding. The human essays come from the BAWE plain-text export, which keeps structural tags such as `<heading>`, `<fnote>`, `<list>`, `<figure>` and `<quote>`. About 88 percent of the human essays in my sample contain at least one of these tags. None of the AI essays do, because Llama writes prose, not corpus markup. A rule as simple as "if the text contains a tag, call it human" reaches 92.5 percent accuracy on the test set on its own, with no understanding of language at all. The AI side has the mirror version of the same problem: Llama sometimes writes markdown (bold, headings, bulleted and numbered lists) that human plain text never has. Both of these are artefacts of how the two halves of the corpus were produced, not differences in writing, and they sit right at the start of the text where the model's 512-token window sees them. This is the shortcut behind a large part of the perfect score.

Does anything real remain once the artefact is removed? Yes. I wrote a normaliser (`src/detection/text_normalize.py`) that strips the BAWE tags from the human side and the markdown from the AI side and flattens the layout, then rebuilt the corpus from the cleaned text. On the cleaned text a simple TF-IDF and logistic-regression model still separates the two classes at 100 percent on the test set. I treat that 100 percent with some caution, because one residual markup token ("`fnote`") survived the cleaning into the full model's

vocabulary. I traced it: the cleaning strips the bracketed tags, but 115 of the 640 AI essays contain the bare word "fnote", echoed by the generator from keyword prompts that were extracted from the human source text. So in the cleaned corpus "fnote" flips from a human marker to a weak AI one; it is a generation artefact rather than leftover markup, it does not touch the function-word or stylometric results, and stripping the bare token from both sides is queued for the next retraining pass. The cleanest evidence is therefore the next result, not this one. A model that is allowed to use only function words (the, and, because, therefore, and the like, which carry no topic information at all, and no markup) still reaches 99.5 percent. Topic cannot explain that, and neither can a leftover tag. The thing that remains after the markup is gone is writing style.

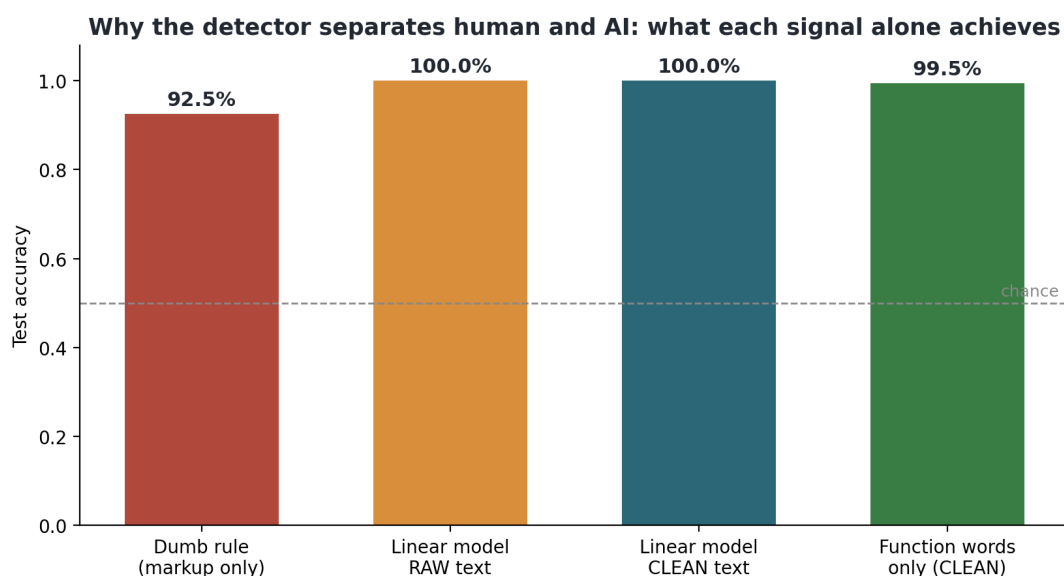


Figure 3.1: What each signal achieves on its own. A markup-only rule already scores 92.5 percent on the raw text, which is the size of the artefact. After cleaning, a simple model still separates the classes, and a function-words-only model still reaches 99.5 percent, so the residual signal is writing style.

What is the model keying on? The cleaned linear model is interpretable, so I can read off the words that push each way. The AI-leaning terms are the familiar large-model register: "in conclusion", "nuanced", "essential", "highlights", "insights", "complex", "significant". The human-leaning terms are blunter connectives: "therefore", "because", "so", "thus", "very". That matches what the stylometric features already showed, that human writing here varies more and the AI writing is smoother and more uniform.

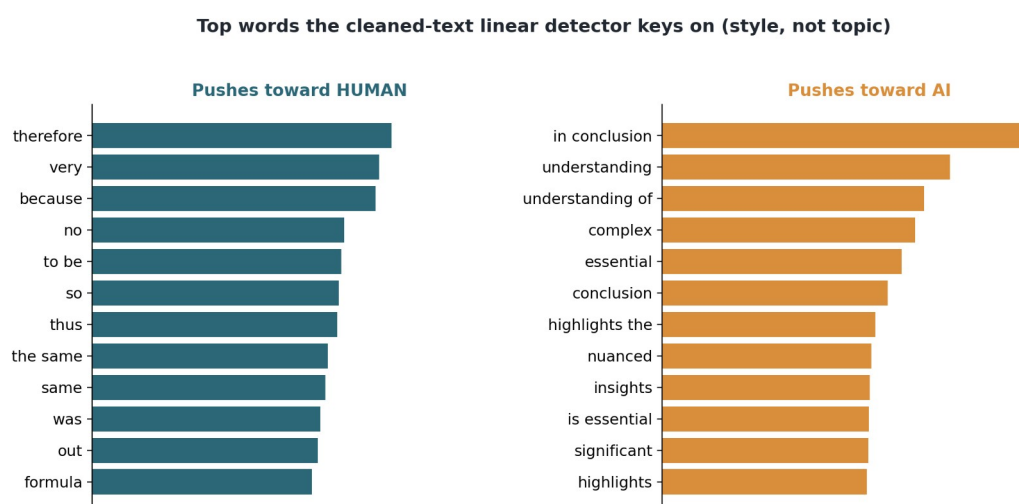


Figure 3.2: The words the cleaned-text model keys on. The AI side is the familiar large-model register; the human side is blunter argument connectives. These are the seeds of the explainability layer.

3.6 The honest headline

After removing the artefact I retrained DeBERTa on the cleaned corpus. The score moved off the ceiling, which is what I wanted: test accuracy 0.99, precision 0.98, recall 1.00, F1 0.990, with a confusion matrix of $\begin{bmatrix} 98 & 2 \\ 0 & 100 \end{bmatrix}$. Two human essays are flagged as AI and no AI essay is missed. The test set is small (100 human and 100 AI essays), so I put a number on that uncertainty: a 95% bootstrap confidence interval on the F1 is about $[0.97, 1.00]$ (`src/evaluation/confidence.py`, Figure 3.5). One extra misclassification moves the F1 by roughly 0.005, so the headline should be read as "about 0.99" rather than an exact value. The full-document linear probe still scores 1.00 on the same cleaned text, while DeBERTa makes two mistakes, which fits the fact that the transformer only reads the opening 512 tokens while the linear model sees the whole essay.

On fairness, the human false-positive rate is 2 of 50 native writers (0.04) and 0 of 50 non-native writers (0.00). With only 50 humans per group, one essay is worth 0.02, so this is not enough to establish a fairness gap, and I treat it as indicative only. The stronger fairness evidence is the cross-domain result in Chapter 6, where the human false-positive rate is measured on far more text and shows the real bias mechanism.

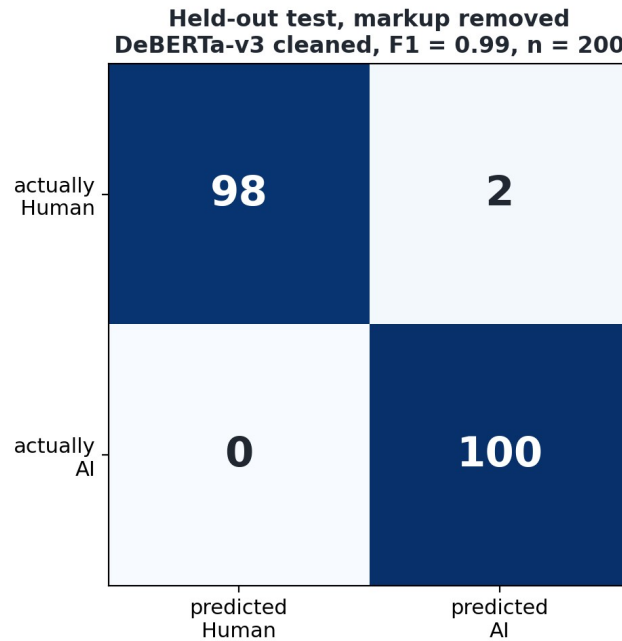


Figure 3.3: DeBERTa on the held-out test set after the markup was removed. Two human essays are flagged as AI and no AI essay is missed (F1 0.990).

RoBERTa, trained the same way on the cleaned corpus, lands in the same place: F1 0.995, a confusion matrix of $[[99, 1], [0, 100]]$, native false-positive rate 0.02. Its only difference from DeBERTa is one human essay flipping, and the two confidence intervals overlap almost completely (Figure 3.5), so on this 200-essay test set 0.990 and 0.995 are statistically indistinguishable and I do not read anything into the gap. What is meaningful is that both architectures land in the same regime: high accuracy, and every mistake is a human essay flagged as AI rather than an AI essay missed. That shared one-sided error pattern, not the near-identical scores, is the sign that the residual signal is a real style difference and not a quirk of one model.

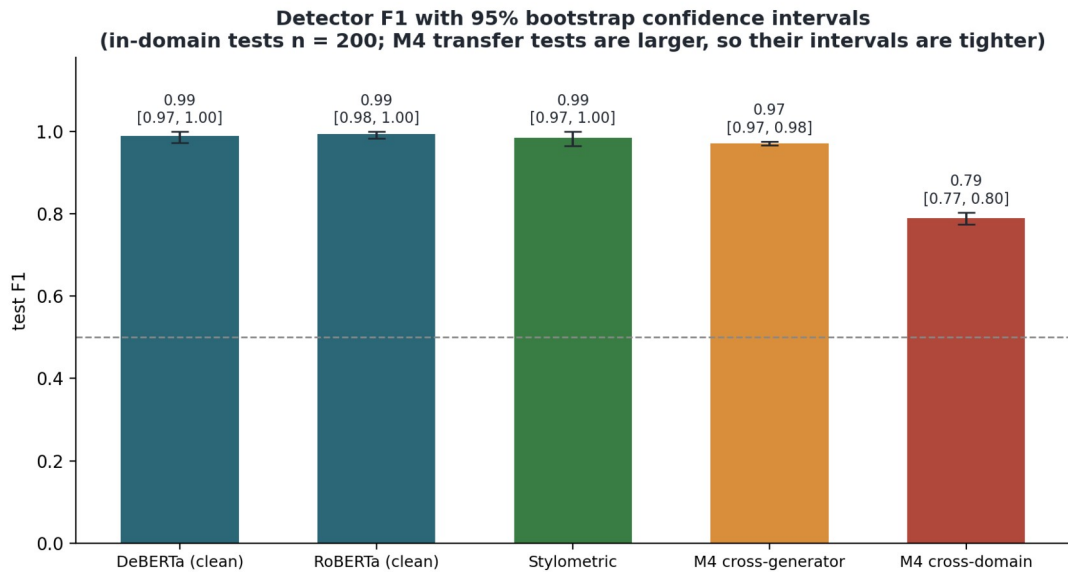


Figure 3.5: Test F1 with 95% bootstrap confidence intervals. The in-domain detectors (DeBERTa, RoBERTa, stylometric) have heavily overlapping intervals on $n=200$, so their differences are within sampling noise; the M4 transfer results have tight intervals on much larger samples.

So the cleaned F1 of 0.990 is the number I report, and the raw 1.000 is kept only to show how large the artefact was. Either way the in-domain task is easy, which is consistent with the literature: separating one known generator from human writing in one domain is close to a solved problem. That is not the contribution of this project. The contribution is the explanation behind each decision and, after this stage, robustness to the harder settings that matter in practice, which are other generators, paraphrased AI text, and submissions that mix human and AI writing. I expect the score to drop there, and that drop is where the real research is.

3.7 Why the score is still high after cleaning

A fair question is why a cleaned score of 0.99 is not itself suspicious. I dug into this (`src/detection/why_high.py`) so I can answer it rather than wave it away. The score is high because the task as I have built it is the easy version of the problem, and several separate signals reinforce each other.

The biggest reason is that all of my AI text comes from one model with one prompt. The AI class is really "Llama 3.1 writing the way I asked it to", so it has a consistent style. When I measure how similar essays are to each other in style space, the AI essays are more alike (average similarity 0.60) than the human essays (0.56), because the humans are a crowd of different people and the AI is one voice repeated 640 times. Separating one consistent voice

from a varied crowd is not hard, and the detection literature treats single-generator, in-domain detection as close to solved.

On top of that sit several smaller tells that all point the same way. The vocabulary differs (the AI register I described above). The spelling differs: the human essays in my sample are in British English (about 2.3 British spellings per thousand words) and Llama defaults to American (about 2.4 American spellings per thousand words). This one is real but shallow, and I am honest that it is a generator-locale artefact rather than deep evidence of AI authorship; it would shrink if I prompted the model to write British English, which is a fix I will try. Formality differs too: the human essays use about three times as many contractions. None of these on its own is decisive, but they stack, and a 2D picture of the essays in pure style space (Figure 3.4, function words only) shows two clouds that barely touch. So 0.99 is the expected result for an easy setting, not a hidden error.

I checked how much of this is the shallow locale tell rather than deep style (`src/detection/style_locale_control.py`). When I remove every British and American spelling variant and every contraction from the text, which changes 1,251 of the 1,280 essays, the function-words-only accuracy does not fall: it goes from 0.995 to 1.00, and the full TF-IDF model stays at 1.00. So the locale and formality tells are real but they are not what the separation rests on; the distributional style signal is there with or without them. I keep the locale point as an honest caveat and a planned fix, not as the explanation for the score.

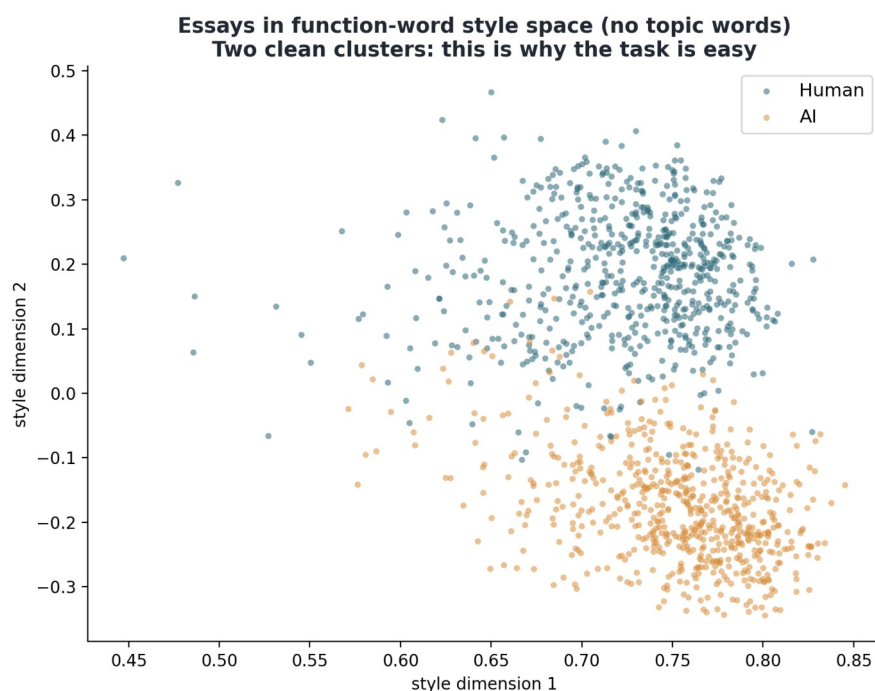


Figure 3.4: Every essay placed by its function-word style alone, with all topic words removed. Human and AI fall into two separate clouds, which is why a single generator in one domain is easy to separate.

The closest call is reassuring rather than worrying. The human essay the model rates most AI-like scores 0.43 on the AI scale, so it is still correctly called human, and the borderline cases are a small mix of native and non-native writers rather than a pile of non-native essays, which is the bias I was watching for.

3.8 Where this chapter's threads are picked up

The audit also seeds the explainability layer. The system can already point at the words that drove a decision, which is the first concrete version of the defensible evidence the whole project is built around. The threads this chapter opens are closed later in the document: the attribution methods (Integrated Gradients, SHAP and attention) are built and faithfulness-tested in Chapter 5, the M4 multi-generator and cross-domain tests are Chapter 6, and the stylometric features are fused with the transformer into the hybrid detector in Section 6.7.

Chapter 4: Implementation

4.1 Environment and reproducibility

The whole project runs on my own laptop. After Meeting 3 it became clear that no ATU HPC or cloud would be available, so every part has to fit a single RTX 4060 with 8 GB of VRAM, or fall back to the CPU. That constraint shaped a lot of the choices below, and it is the reason the models are base-sized rather than large.

The stack is Python on Windows, with PyTorch and the Hugging Face libraries for the transformer work, spaCy for the linguistic features, and scikit-learn for the simpler baselines and the audit probes. To keep runs repeatable I set a single random seed (42) everywhere that randomness enters: the sampling, the train and test split, and the model training. Dependency versions are pinned, and the work is committed to a local git repository in small steps with clear messages, so any result can be traced back to the exact code that produced it. Long jobs (essay generation, model training) are launched as detached background processes rather than tied to an open terminal session, so they keep running if the session that started them closes. This sounds like a small detail, but it cost me two failed generation runs before I worked it out, and it is now the standard way I start anything that takes more than a few minutes.

4.2 Data acquisition and cleaning

The human side of the corpus is the British Academic Written English (BAWE) collection (Alsop and Nesi, 2009), which I downloaded from the Oxford Text Archive. A first script (`src/data/explore_bawe.py`) reads the holdings spreadsheet, prints a summary, and cross-checks the recorded word counts against a plain count of the text files so I know the metadata is trustworthy. A second script (`src/data/clean_bawe.py`) produces a cleaned metadata table, dropping a row that was labelled twice. The cleaned table is what every later step samples from.

4.3 Sampling

The sample is drawn by `src/data/build_sample.py`. Rather than sample by named discipline, I stratify evenly across the four broad disciplinary groups and balance native against non-native English writers, so that I can later test whether the detector is unfair to non-native

writers. A per-student cap stops any one person's style from dominating, and the train, validation and test split is made at the level of the student, not the essay, so the same writer never appears on both sides of the split. The result is 640 human essays with a versioned manifest (`bawe_human_sample_manifest.csv`) that records, for each essay, its group, first-language status, and split. The reasoning behind the sizes is written up in `dissertation/sample_design.md`.

4.4 Generating the matched AI essays

The AI half of the corpus is built by `src/generation/generate_ai_essays.py`. For each human essay it generates one AI essay on the same topic and at the same target length, using Llama 3.1 8B running locally through Ollama. The length and topic matching is the most important part: if the AI essays were systematically longer or shorter, or drifted onto different subjects, the detector could separate the classes for the wrong reason and the whole corpus would be invalid.

Two things make the matching work. First, the prompt is anchored on keywords pulled from the human source text, not just the essay title, after an early test produced an off-topic essay from a title alone. Second, the generator runs a short continuation loop: if the first draft falls short of the target length it asks the model to continue, up to a few rounds, so the AI essay lands close to its human counterpart. The script is resumable, so a run that stops can be restarted and it skips what is already done, and it keeps the machine awake while it runs. It writes each essay to disk along with a metadata row recording the target and actual length, the number of rounds, and the time taken.

Generation of all 640 essays completed locally with no failures. A validation script (`src/generation/check_ai_corpus.py`) confirmed the match: the correlation between each AI essay and its human source length is about 0.98, the mean length ratio is about 1.05, and over 99 percent of AI essays are within 20 percent of their source length. A keyword spot-check confirmed the AI essays stayed on the same topics. So the two halves differ in writing, not in length or subject, which is exactly the property the detector needs.

4.5 Building the labelled corpus

`src/detection/build_detection_corpus.py` pairs each human essay (label 0) with its matched AI essay (label 1), carries the split and metadata across from the manifest, and

writes a single table of 1,280 rows. The same script has a cleaning mode (described in Section 4.7) that strips markup before writing, so the raw and cleaned corpora can be compared directly.

4.6 The detector

The detector is fine-tuned by `src/detection/train_detector.py`. It uses the Hugging Face Trainer to fine-tune a transformer to classify human against AI, with DeBERTa-v3-base as the primary model and RoBERTa-base as a comparison. To fit the 8 GB card I keep the per-step batch small and use gradient accumulation to reach a sensible effective batch, with mixed precision on the GPU. The script reports accuracy, precision, recall, F1 and the confusion matrix, and on top of that it breaks out the false-positive rate for native and non-native writers separately, which is the seed of the fairness analysis. The stylometric feature extractor (`src/detection/stylometric.py`) computes the linguistic features (sentence-length variation, vocabulary richness, part-of-speech mix, and so on); those features are fused with the transformer into the hybrid detector in Section 6.7 (`src/detection/hybrid_fusion.py`).

4.7 The audit and the cleaning step

The first detector scored a perfect 100 percent, which I did not trust. The audit that followed is implemented in three small scripts, and they are part of the contribution, not throwaway checks.

`src/detection/audit_detector.py` runs the diagnostic battery: it confirms there is no student or duplicate-text leakage across the splits, measures how much of the score a markup-only rule can reach on its own, and trains simple interpretable baselines on raw and cleaned text, including a function-words-only model that removes all topic information. `src/detection/text_normalize.py` is the cleaning function that the audit relies on: it strips the BAWE export tags from the human text and the markdown from the AI text, and flattens the layout, so that any remaining difference is writing style rather than formatting. `src/detection/why_high.py` then explains why the cleaned score stays high, by measuring locale spelling, contraction use, and how tightly the AI essays cluster in style space, and by drawing the two classes in a two-dimensional style projection.

The finding, covered in Chapter 3, was that the original human text carried structural tags that no AI essay had, and that this shortcut accounted for a large part of the perfect score. The cleaning step removes it, and the detector is then retrained on the cleaned corpus to give the honest headline figure.

4.8 Engineering notes and lessons

Two practical lessons are worth recording for the methodology. The first is the value of launching long jobs as detached processes, which is what finally made the overnight generation reliable. The second is the markup artefact itself: it is a clean example of why a strong result has to be stress-tested before it is believed, and the fact that the audit is scripted means anyone can rerun it and see the same thing. Both points feed directly into how I will build and check the remaining components.

Chapter 5: Explaining the detector's decisions

5.1 Why this matters for the project

The whole point of the project is that a detector should hand a lecturer defensible evidence, not just a percentage. So the detector is only half the story. The other half is being able to say why a piece of text was flagged, and being honest about whether that explanation is real. This chapter is the first version of the explainability layer. It does two things: it shows which parts of an essay drove the decision, and then it tests whether that explanation is faithful, which means whether the words it points to are actually the words the model used.

5.2 Token attributions with Integrated Gradients

For the word-level view I used Integrated Gradients, a standard attribution method, through the Captum library (`src/explainability/integrated_gradients.py`). It works on the detector's word embeddings and measures how much each token pushes the decision toward the AI class, compared to a neutral baseline where the content is removed. I attribute everything to the AI class so the sign is consistent: a positive score pushes toward AI, a negative score pushes toward human.

Figure 5.1 shows this for a matched pair, the AI and human versions of the same essay. Two things stand out. The detector is confident and in the right direction for both (the AI essay scores 1.00 on the AI scale, the human essay 0.01). And the tokens that carry the most weight are not topic words. They are punctuation and very common words and word-fragments: full stops, commas, a dash, words like "have", "be", "can", "similarly". That fits everything from the audit. The model is reading style and rhythm, the small machinery of how a sentence is put together, rather than what the essay is about.

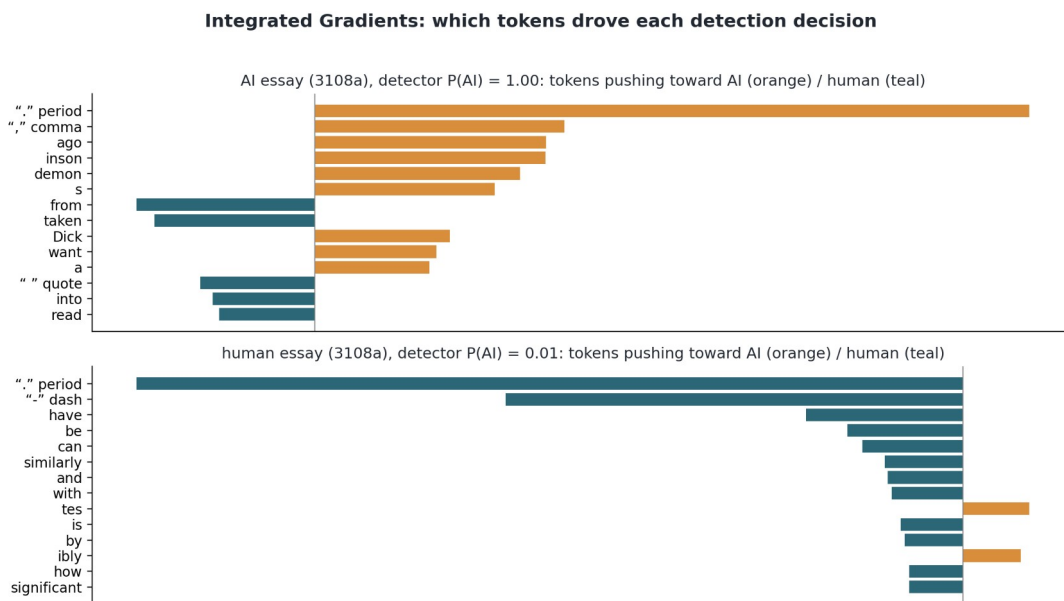


Figure 5.1: Integrated Gradients token attributions for a matched AI and human essay. Orange pushes toward AI, teal toward human. The strongest tokens are punctuation and common words, not topic words, so even at the word level the detector is reading style.

5.3 Are the highlights honest? A faithfulness test

A highlight is only useful if it is faithful. I tested this by ablation (the same script). For a sample of test essays I ranked the tokens by their attribution and then asked two questions. First, comprehensiveness: if I remove the top tokens, does the detector's confidence fall, and does it fall more than if I remove the same number of random tokens? Second, sufficiency: if I keep only the top tokens and hide the rest, does the confidence hold up?

Figure 5.2 shows the comprehensiveness result as I remove more and more tokens. Removing the top-ranked tokens does hurt the detector more than removing random ones, but only by a little, and only once a fair number are removed. This comprehensiveness sweep is the load-bearing, honest part of the test. I also ran a sufficiency check (keep only the top tokens, hide the rest), but with at most a few dozen of 256 tokens kept the model sees almost nothing and sits at a coin-flip for every k , so that flat result is an empty-input artefact of the design rather than a finding, and I do not rely on it. The comprehensiveness result on its own already says what matters: no small set of words is enough to carry the decision.

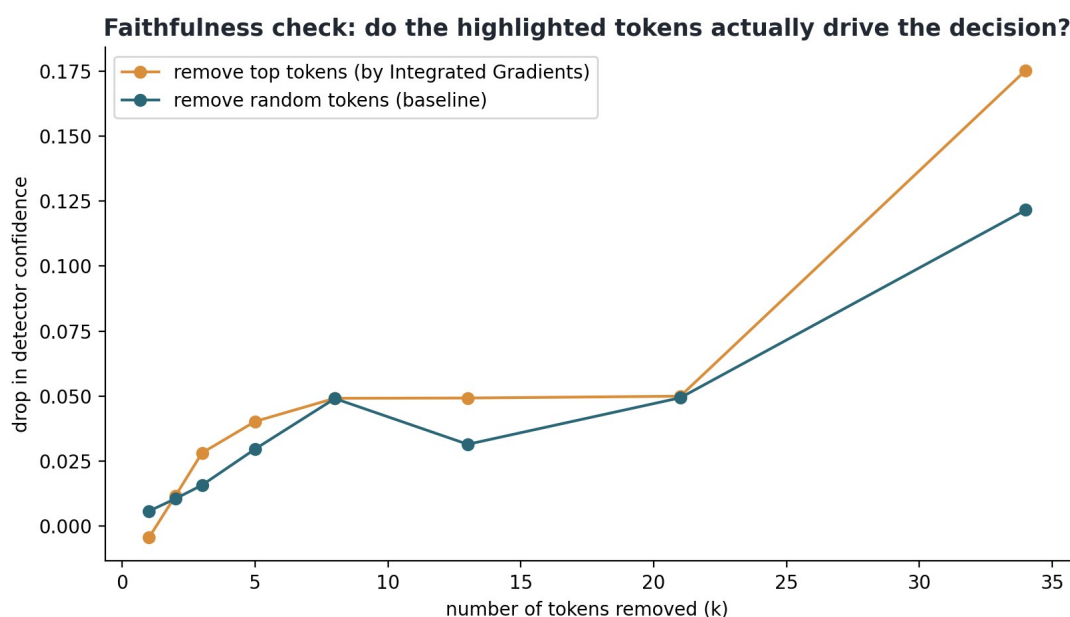


Figure 5.2: Faithfulness by ablation. Removing the top attributed tokens (orange) lowers detector confidence only slightly more than removing random tokens (teal). Keeping only the top tokens collapses the prediction to chance. The signal is spread across the whole essay.

5.4 What this tells me: the signal is diffuse

This is an honest and useful negative result. The reason a few words cannot explain the decision is that the difference between human and AI writing here is spread across the whole essay, in the ordinary words and the punctuation, not concentrated in a handful of give-away terms. It is the same thing the audit found from the other direction: a model using only function words still separated the classes at 99.5 percent, and the essays form two clean clouds in pure style space. A diffuse signal is hard to summarise with a word-level highlight, so for this detector a per-word heatmap is a weak explanation on its own.

The explanation that is faithful for this detector is the feature-level one. The interpretable linear view from the audit (Chapter 3) names the style markers that separate the classes, the large-model register on one side and the blunter connectives on the other, and that view is backed by the function-words-only result rather than contradicted by it. So the defensible, lecturer-facing explanation is the style and feature picture, supported by the source-grounded verification questions that come later in the pipeline, not a single highlighted sentence.

5.5 The faithful explanation: stylometric features and SHAP

To make the feature-level explanation concrete I built a detector from hand-crafted style features alone, with no transformer: sentence-length variation and burstiness, vocabulary

richness, word length, punctuation, and the part-of-speech mix (`src/explainability/shap_stylometric.py`). Trained on the same student-level splits, this transparent model reaches an F1 of 0.985 on the test set (95% bootstrap confidence interval about [0.97, 1.00], and identical, 0.985, across five training seeds, so it is stable). Two honest caveats. The comparison with the transformer's 0.99 is not quite like for like: the feature model reads the whole essay while DeBERTa reads only the first 512 tokens, which helps the feature model, so I read this as "competitive" rather than "equal". And the false-positive rate of 0.02 for native and non-native writers is one essay in fifty each, too small to claim fairness; the cross-domain result in Chapter 6 is the real fairness evidence. Two of the features (type-token ratio and the rare-word ratio) are length-sensitive, but since the essay lengths are matched across the classes this carries little class signal; the length-robust vocabulary measure behaves the same way. Perplexity, one of the strongest stylometric signals in the literature, needs a GPU pass and joins the feature set when the hybrid is assembled in Section 6.7, where it proves to be the strongest single feature. Even so, the headline holds: the signal is captured almost as well by a handful of interpretable features as by a large model.

Because the model is built from named features, I can explain it faithfully with SHAP, which attributes each decision to the features that drove it. Figure 5.3 shows the picture across the test set. Longer words and a denser use of auxiliary verbs push a text toward AI, while more sentence-length variation, richer vocabulary, and more rare one-off words push it toward human. This is the explanation the project actually needs: a lecturer can be told that a piece was flagged because its sentences are unusually uniform, its words longer, and its vocabulary less varied than a typical student's, and that account is faithful because the model literally uses those features. It is the opposite of the black-box percentage the project set out to replace.

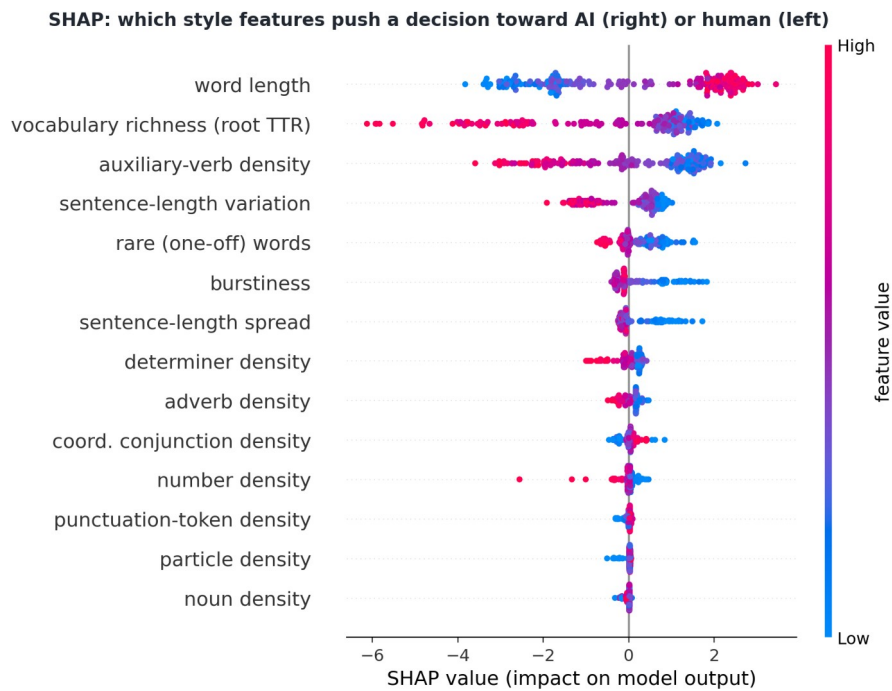


Figure 5.3: SHAP on the stylometric detector. Each dot is an essay; position shows how much a feature pushed the decision toward AI (right) or human (left), and colour shows the feature value. Longer words push toward AI; richer vocabulary and more varied sentence length push toward human.

This stylometric model is also the non-transformer half of the planned hybrid detector, so this step serves both the explainability layer and the detector itself.

5.6 Attention, the third method, on the same yardstick

The explainability component names three methods, and attention visualisation is the third (`src/explainability/attention_viz.py`). The view is the classic one: how strongly the classifier position attends to each token in the final layer, averaged over heads, which is often presented as "what the model looked at". On the matched pair from Section 5.2 it produces a plausible-looking picture (Figure 5.4): in the AI essay the most-attended token is an em dash, followed by content words like "extract" and "poetry", while in the human essay the attention mass sits on ordinary function words like "the", "as" and "is". The em dash detail is a pleasing echo of the stylometric story, since dash-heavy punctuation is one of the register tells the feature model also picks up.

Plausible-looking is not faithful, though, and attention weights have a reputation for being poor guides to what actually drives a decision. So the same ablation protocol as Section 5.3 was applied, same test sample, same seed, same k-sweep, making all three methods comparable on one yardstick. Attention lands almost exactly where Integrated Gradients does

(Figure 5.5): removing its top 34 tokens drops confidence by 0.17 against 0.12 for random tokens, a ratio of 1.43 to IG's 1.44, and keeping only its top tokens leaves the detector near a coin flip (sufficiency around 0.53), just as it does for IG. At small k attention is even marginally ahead of IG, but neither method's top tokens are anywhere near sufficient to carry the decision.

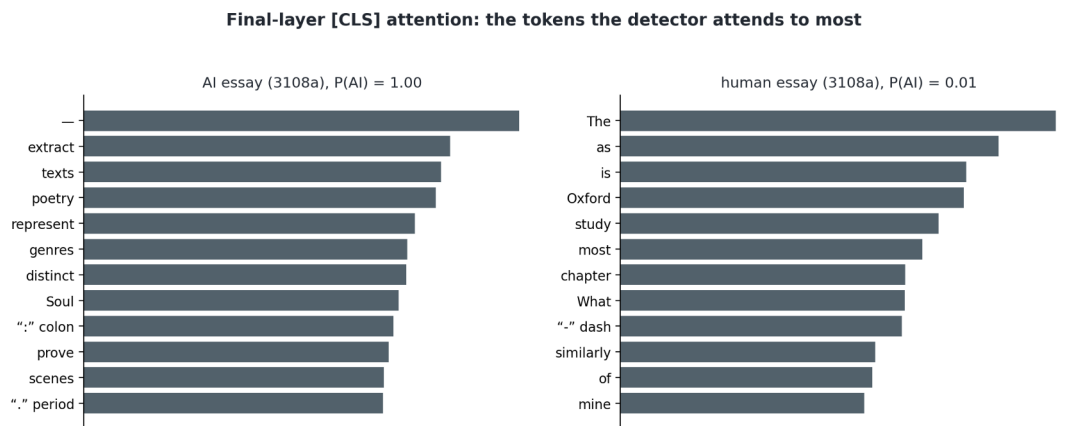


Figure 5.4: Final-layer attention from the classification position for the same matched pair as Figure 5.1. The picture looks interpretable, and in the AI essay the single most-attended token is an em dash, but the faithfulness test is what decides whether it can be trusted.

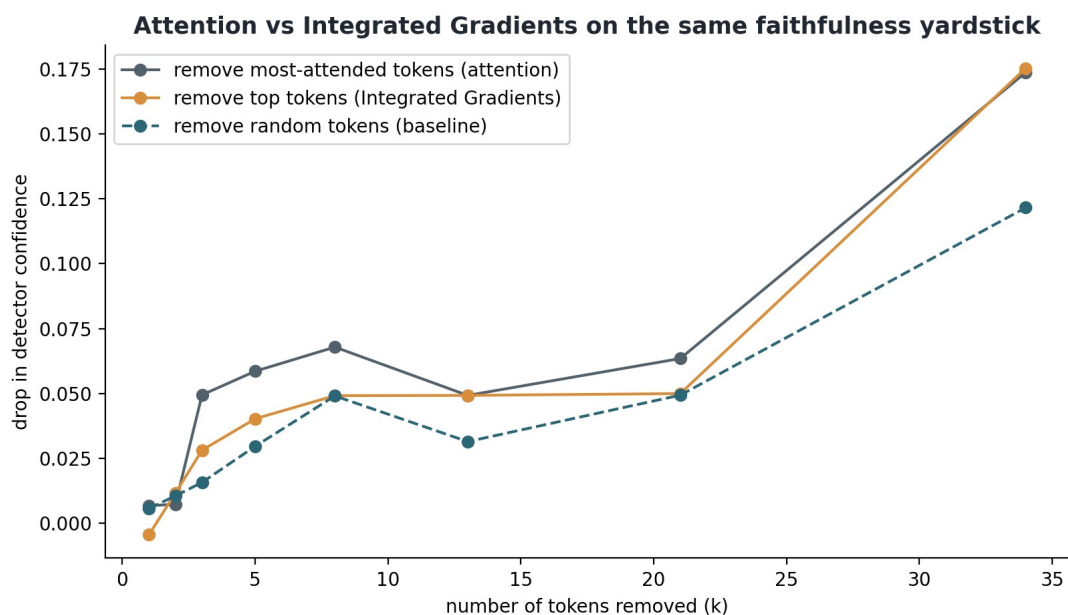


Figure 5.5: Attention and Integrated Gradients on the identical faithfulness protocol. The two token-level methods are nearly indistinguishable, both only modestly above the random-removal baseline, which is the diffuse-signal finding again from a third angle.

The conclusion triangulates rather than changes. Three methods now sit on one measured yardstick: attention and Integrated Gradients agree with each other and are both weakly faithful, because the style signal is spread across many ordinary tokens rather than

concentrated in a few, while SHAP over the named stylometric features passes its ablation test and remains the explanation shown to lecturers. Having the third method measured matters for the argument: the weakness is not a quirk of one attribution technique, it is a property of token-level explanation on this task.

5.7 What follows for explainability

Three things follow. First, the stylometric model and the transformer are fused into the hybrid detector in Section 6.7, with perplexity added to the feature set. Second, keep the token attributions as a supporting visual, useful for showing a lecturer roughly where the model looked, but always paired with the honest caveat from the faithfulness test, with the SHAP feature view as the primary explanation. Third, carry the faithfulness check forward as a standard step: any explanation the system shows should be one I have tested, not one I have assumed.

Chapter 6: Robustness, transfer to unseen generators and domains

6.1 The question

The in-domain detector scores about 0.99, but that is on the easy setting: one generator (Llama) and one kind of text (student essays). The honest question is what happens when neither holds. To test it I took the trained detector, with no retraining or adaptation, and ran it on the M4 benchmark (SemEval-2024 Task 8), which has many generators and many domains that the detector never saw. I split this into two tests so I do not accidentally call the easy half "robust".

A note on sources, so the two tests are not confused. Test A uses the OUTFOX essay set that ships inside the M4 release: human essays against six generators. Test B uses the M4 / SemEval-2024 Task 8 monolingual data across five web and academic domains. They are different corpora with different generator sets, which I chose deliberately: A isolates a change of generator while holding the text type fixed (essays), and B changes the domain. I name them separately rather than calling both simply "M4".

6.2 Test A: transfer to unseen generators (still essays)

The first test uses the OUTFOX essay set, where humans and six different models (GPT-4, ChatGPT, Cohere, BLOOMz, Dolly, and davinci) all wrote essays. This keeps the kind of text the same as training and only changes the generator. The detector held up well: an F1 of 0.97 (95% confidence interval [0.97, 0.98], on about 4,800 essays so the interval is tight), and it caught every generator at between 96 and 100 percent (Figure 6.1), with a human false-positive rate of about 5 percent. This is a useful and slightly surprising finding. A detector trained only on Llama still recognises text from quite different models. The reason is that current large models share a lot of the same smooth, uniform style, so the AI fingerprint is largely generator-agnostic, at least on essays.

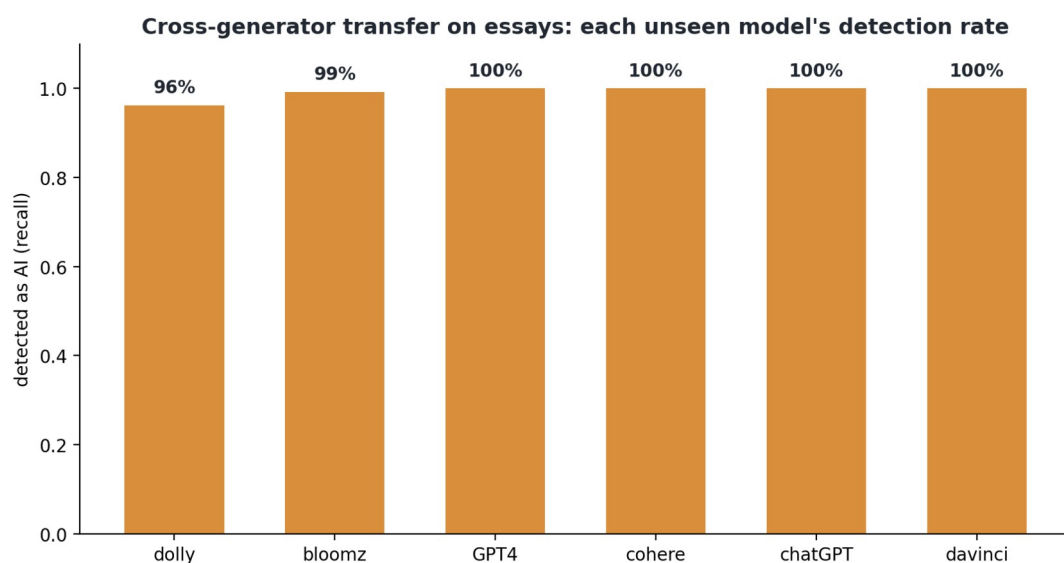


Figure 6.1: Transfer to unseen generators on essays. A detector trained only on Llama still flags GPT-4, ChatGPT, Cohere, BLOOMz, Dolly and davinci at 96 to 100 percent.

6.3 Test B: transfer to unseen domains

The second test is the hard one, on the M4 monolingual data. Here the human text is not essays at all: it is Reddit posts, WikiHow articles, arXiv abstracts, Wikipedia, and peer-review text, against four generators. The overall F1 falls to 0.79 (95% confidence interval [0.77, 0.80], tight because the sample is large), and the way it fails is the important part. The generator set here differs from Test A, so the headline drop mixes a change of domain with a change of generators. The load-bearing evidence is therefore not the combined F1 but the human side of the result, because a false-positive rate measured on human text cannot be a generator artefact. The detector still catches the machine text well in every domain (86 to 98 percent). What breaks is the human side: it wrongly flags genuine human writing as AI at high rates on the more formal domains, 79 percent on arXiv abstracts, about 40 percent on Wikipedia and WikiHow, 30 percent on peer-review, and 23 percent on Reddit (Figure 6.3). That human-side failure is what carries the argument.

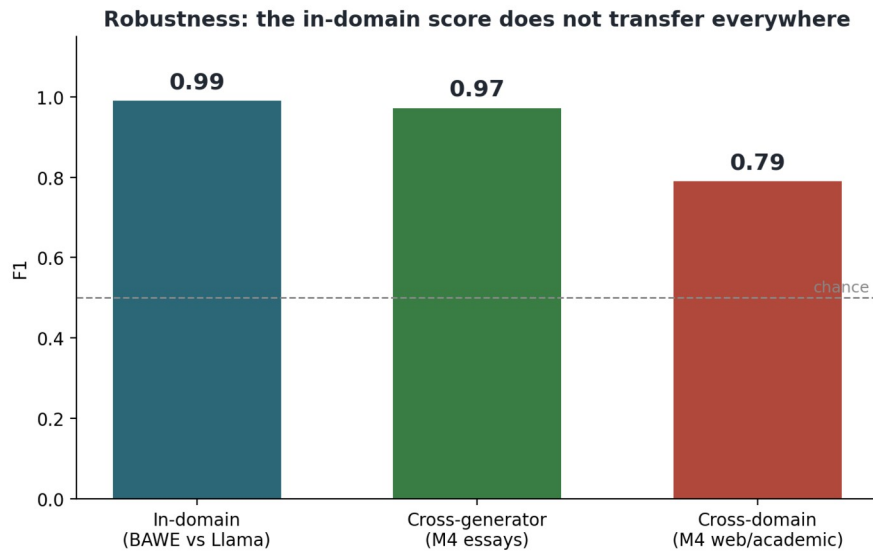


Figure 6.2: The in-domain score does not transfer everywhere. F1 stays high across generators on essays but falls on out-of-domain human and AI text.

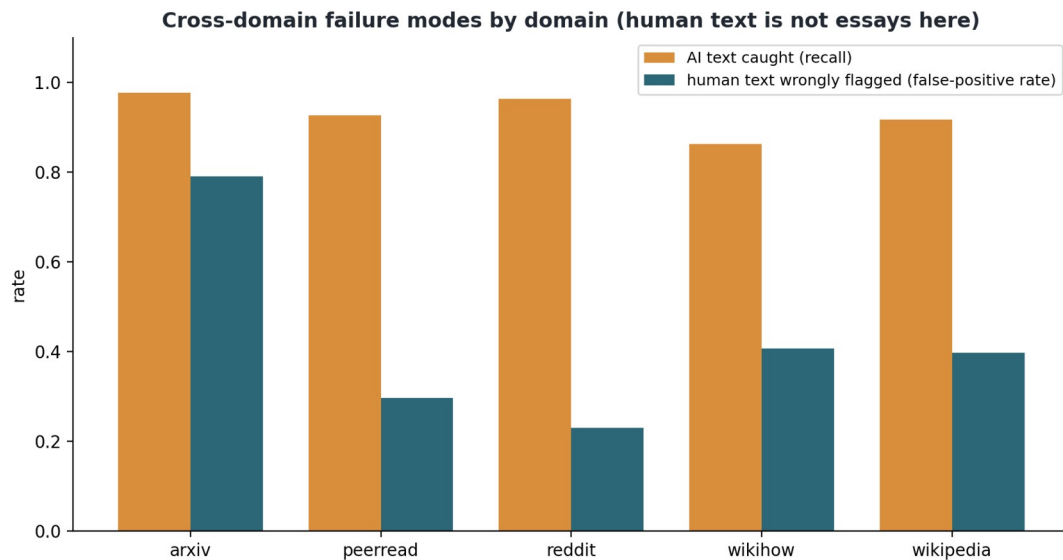


Figure 6.3: Cross-domain failure modes. The detector still catches AI text (orange) but wrongly flags formal human text as AI (teal), worst on arXiv abstracts.

6.4 What this means

The picture is clear and it matters for the project. The detector is robust across AI models but fragile across domains, and the fragility is not random. It learned that "human" looks like a student essay, so when it meets human writing that is more formal or more technical, an arXiv abstract for example, it calls it AI. In other words, the out-of-domain failure is a false-accusation failure, exactly the harm the project is trying to avoid.

This connects directly to the fairness concern from the start of the project. A detector trained on one narrow idea of human writing will misjudge writers whose style sits outside that idea.

The arXiv result is the clearest warning: formal, dense, careful human prose reads as machine to this model. The same mechanism is what makes these tools risky for non-native writers, whose style also differs from the training norm. It is also the strongest argument for why the pipeline does not stop at a score. A number that is confidently wrong on a whole domain is dangerous on its own; the verification questions are what turn a flag into something a lecturer can check rather than act on blindly.

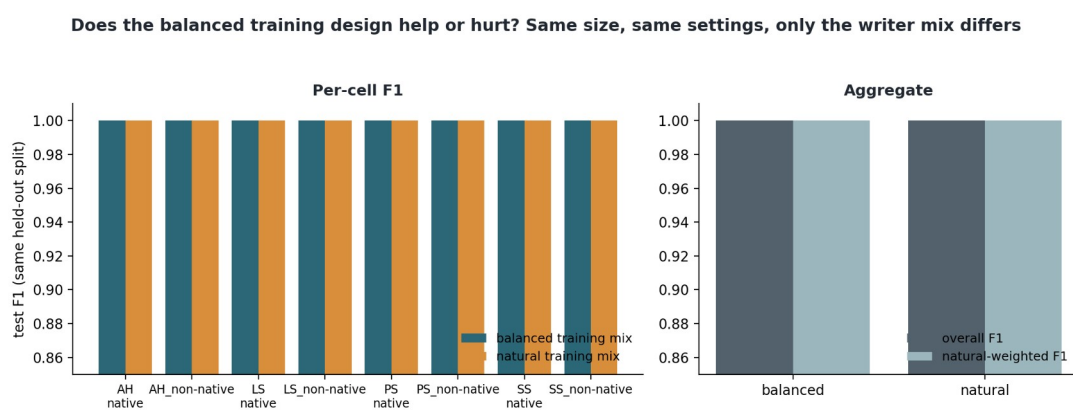
6.5 Caveats and next steps

This is a zero-shot test with no adaptation, so it is a lower bound on what is achievable: a detector trained on diverse human text and several generators would almost certainly do better across domains, and that is a clear next experiment (train on a slice of M4 and re-test). The in-domain "human" being essays is the root of the false positives, so the fix is more varied human data and per-domain calibration. The remaining robustness gaps to test are paraphrased and "humanised" AI text, and documents that mix human and AI writing. For the project as it stands, the honest headline is that the detector is good at spotting AI essays from many models, and not safe to point at human writing from a domain it was not trained on.

6.6 Does the balanced corpus design distort? The natural-distribution control

One design worry was agreed with my supervisor early (Meeting 2) and stayed open until now. The detection corpus is deliberately balanced, equal essays per disciplinary-group-by-native cell, but real submissions are nothing like that: in the full BAWE corpus, native Arts and Humanities writers supply about 21 percent of essays and non-native AH writers about 4 percent. If the detector's strong results depended on that artificial structure, the whole evaluation would be over-fitted to a corpus shape that never occurs in practice. The control is two detectors trained in parallel from the same cleaned corpus with identical hyperparameters, seeds and size (247 human essays each, plus their matched AI twins), differing only in writer mix: one balanced (31 per cell), one matching full BAWE's natural skew (53 native AH essays down to 10 non-native AH). Both are evaluated on the same untouched test split, per cell, reweighted to the natural proportions, and with the fairness read (`src/evaluation/balanced_vs_natural.py`, `outputs/balanced_vs_natural.json`).

The answer is that the balancing is not load-bearing (Figure 6.4). The two models score identically, F1 1.000 overall, in every cell, and under natural reweighting, and they make the same prediction on every one of the 200 test essays (zero disagreements, McNemar $p = 1.0$), with a zero false-positive rate on human essays in both native-language groups. Read against Chapter 3 this is unsurprising: the cleaned corpus is separable on function words alone, so 494 training essays saturate the test set regardless of how their writers are distributed. Two conclusions follow, one reassuring and one honest. The reassurance is that the balanced design did not manufacture the result; a realistically skewed training mix reproduces it exactly, which is what the Meeting 2 guard was for. The honesty is that at this level of separability the comparison has no resolution: whether balancing helps or hurts can only be measured where the task is hard, such as the cross-domain settings of this chapter, and that is where a future version of this control belongs.



† two models made identical predictions on every test essay (disagreements: 0 vs 0, McNemar $p = 1.0$). At this corpus's separability, the training mix is not load-bearing

Figure 6.4: The training-distribution control. Two same-size detectors, one trained on the balanced writer mix and one on BAWE's natural skew, evaluated per cell and in aggregate on the same held-out split. They are indistinguishable everywhere, so the balanced design is not what makes the corpus separable.

6.7 The hybrid detector, and what fusion actually buys

The scope's first component is a hybrid: the transformer combined with the stylometric features, including the GPT-2 perplexity signal (Radford et al., 2019) that Chapter 5 deferred because it needs a GPU pass. Both halves have existed for some time at 0.990 and 0.985 in-domain, so assembling the hybrid could not be about the headline number; the interesting question, given everything this chapter found, is whether the fusion changes the detector's behaviour where it fails. The assembly is simple and disclosed: perplexity joins the feature set, the feature model is retrained, and a logistic regression fuses the two models' probabilities, fitted on the validation split so the test set stays untouched, then applied zero-

shot out of domain (src/detection/hybrid_fusion.py, outputs/hybrid_fusion.json).

In-domain the answer is quick: everything sits at the ceiling. Adding perplexity lifts the feature model from 0.985 to 1.000 on the 200-essay test split, the hybrid scores the same, and with two essays' worth of movement separating all the arms these differences should not be over-read. The detail worth keeping is that perplexity enters the feature model as its strongest single feature, first of twenty-five by mean absolute SHAP, which confirms the literature's regard for it and completes the feature set the scope specified.

Out of domain is where the fusion earns its place (Figure 6.5). On the same cross-domain sample as Section 6.3, the three arms have almost identical F1 (transformer 0.790, hybrid 0.791), but they make completely different errors. The transformer over-flags humans: recall 0.93 at precision 0.69, which is the false-accusation profile. The style-plus-perplexity model is the mirror image, precision 0.82 at recall 0.72. The fusion balances the two (precision 0.80, recall 0.78) and lifts accuracy from 0.753 to 0.794, and the effect on the failure that matters is large: the human false-positive rate falls in every domain, from 0.79 to 0.61 on arXiv abstracts, 0.30 to 0.09 on peer reviews, 0.23 to 0.05 on reddit, 0.41 to 0.10 on wikihow and 0.40 to 0.12 on wikipedia. In four of the five domains false accusations drop by a factor of three to eight; arXiv remains the hard case, better but not fixed. The style half is what does it: hand-crafted features generalise across registers where the transformer's learned representation of "human" stays anchored to student essays.

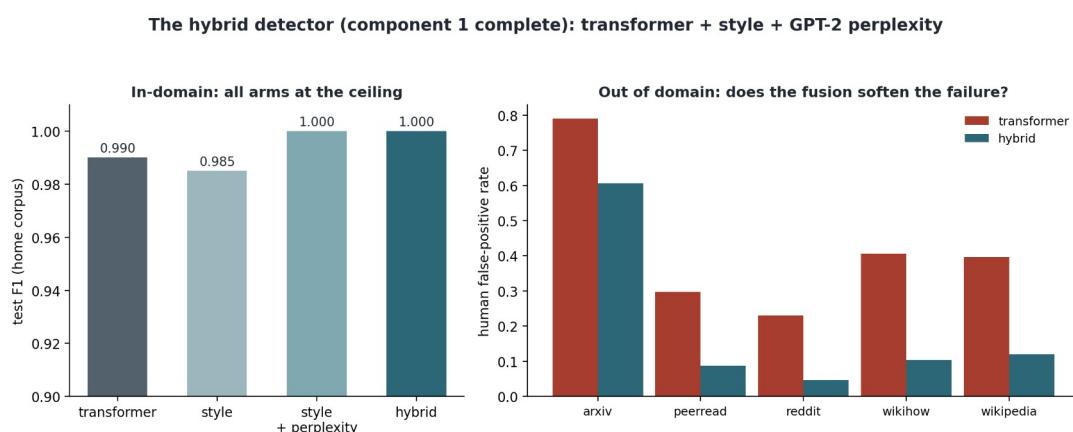


Figure 6.5: The hybrid detector. In-domain (left) every arm is at the ceiling and the differences are within noise. Out of domain (right) the fusion barely moves F1 but changes the error mix: the human false-positive rate, the false-accusation failure of Section 6.3, falls in every domain, by three to eight times in four of the five.

The reading connects back to Section 6.4's mitigation plan. Fusion does not solve domain shift, and the headline F1 barely moves, so this is not a robustness cure. What it does is redistribute the remaining errors away from falsely accusing human writers, which is exactly the direction a deployed system must err in, and it does so for the price of a feature extractor that runs anywhere. Component 1 of the scope is now complete as specified, and the hybrid, not the bare transformer, is the detector the rest of the pipeline should call.

Chapter 7: From a flag to verification questions

7.1 Why this is the point of the project

Detection on its own only produces a flag, and the earlier chapters showed how fragile a flag can be out of domain. The actual contribution of the project is what happens after the flag: turning a suspected submission into something a lecturer can act on fairly, which is a short set of questions, drawn from the student's own claims, that check whether the student understands what they handed in. A student who wrote and understood the work can answer them; a student who did not will struggle. This chapter builds that step: a first thin slice end to end, then the trained components that replace its stand-ins one by one, and the fine-tuning of the local backend.

7.2 How the slice works

The pipeline takes a flagged essay and produces a Verification Interview Guide (`src/question_gen/generate_questions.py`). It runs in four steps.

First, it splits the essay into numbered sentences. Second, it extracts the student's main claims, and for each claim it records the sentence numbers the claim comes from. This is the important design choice for provenance: the model is asked to cite sentence numbers, not to quote the text, so the source is looked up from the real numbered sentences and the model cannot invent a quote. Every claim therefore points at exact lines in the submission. Third, for each claim it generates verification questions that are grounded in the source sentences and written so that they cannot be answered well by someone who only read the claim. Fourth, each question is tagged with a Bloom's cognitive level.

Backends sit behind one interface. This first slice runs on the local open-source model (Llama 3.1 8B through Ollama), which is the basis for Backend B. The commercial Backend A plugs into the same interface, and because the backend is recorded with every guide the two can be compared directly; that comparison, the core commercial-versus-local research question, is carried out in Chapter 8.

7.3 An example

Run on a flagged essay that compares three literary texts, the system pulled out four claims and wrote grounded questions for each. Two of them give the flavour.

For the claim that the three texts use language differently (metaphor and symbolism in the poetry, dramatic intensity in the drama, subtle nuance in the prose), tied to two specific sentences in the submission, it asked: "What specific features of Shakespeare's dialogue lead you to describe it as having dramatic intensity?" and "How does Austen's subtle and nuanced prose style allow her to explore complex social issues in a way that might be less effective with more overt language?" Both need the student to reason about the text, not repeat the claim.

For the claim that the texts share themes of love and personal growth, tied to a single sentence, it asked for an example from one of the texts that links personal growth to relationships, which is a question a student who genuinely engaged with the texts can answer and a student who did not cannot. The full guide, with every claim, its exact source sentences, and the Bloom's levels, is saved as `outputs/verification_guides/3108a_ai.md`.

7.4 What was first-slice, and where each stand-in went

The slice above was deliberately a thin path through the remaining pipeline, with three declared stand-ins, and it is worth recording what they were because the rest of this chapter replaces them one by one. The claim extraction was prompted rather than the trained argument miner planned in the scope; the trained extractor arrives in Section 7.6. The Bloom's tag was a transparent keyword heuristic, replaced by the trained classifier in Section 7.5. And the questions themselves were not yet evaluated; the discrimination simulation that turns them into measured results is the subject of Chapter 8, where the commercial backend also joins the comparison. The one open decision at this point, whether the local backend should be the full Llama 3 8B with QLoRA or a smaller model that fits the laptop, was settled with my supervisor on the evidence of the fit probes and is picked up in Section 7.8.

7.5 The Bloom's classifier: replacing the heuristic

The keyword heuristic above was always a placeholder, and component 5 of the scope is a trained classifier. I fine-tuned BERT-base (Devlin et al., 2019) on the Bloom-labelled subset of EduQG (Hadifar et al., 2022): 903 questions carrying a cognitive level, of which only four levels occur (remember, understand, apply, analyse) in a heavily skewed distribution (660, 114, 110 and 19 examples respectively). Training used class weights against that imbalance, stratified

splits (632 train, 135 validation, 136 test), and six epochs; the transparent keyword heuristic was scored on the same held-out test split as the baseline.

The trained model roughly doubles the heuristic: macro-F1 0.31 against 0.16, accuracy 0.57 against 0.26 (Figure 7.1). The per-class picture is the honest part. BERT is reliable on the majority class (remember, F1 0.76) and moderate on understand (0.38), but it cannot learn apply (0.09) or analyse (0.0) from 110 and 19 examples. Coarsening the same predictions into the standard lower-order versus higher-order split does not rescue the minority side either: higher-order F1 is 0.23 for BERT and 0.17 for the heuristic, and the heuristic's apparently high binary accuracy (0.86) is majority-class bias rather than skill, since predicting lower-order for everything scores about 0.85 on this test set.

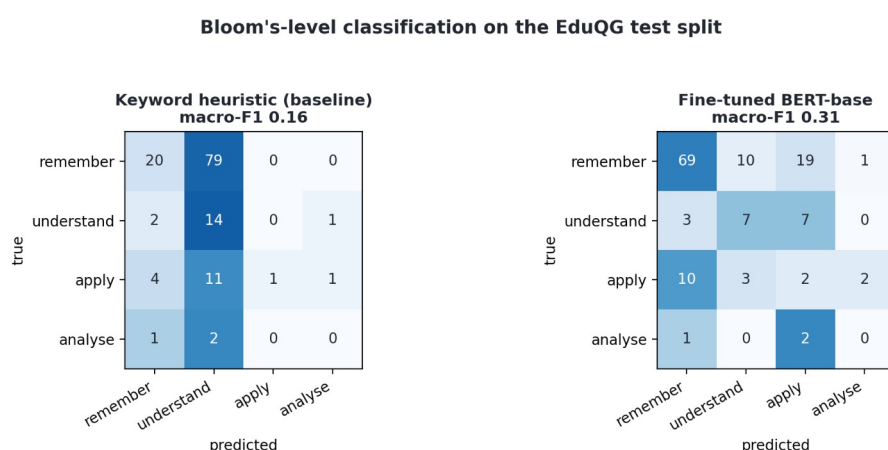


Figure 7.1: Bloom's-level classification on the EduQG test split. The trained BERT-base doubles the keyword heuristic on macro-F1 (0.31 vs 0.16), but neither model can learn the two smallest classes from 110 and 19 examples.

Two conclusions follow. The component works and clearly beats the transparent baseline, so it replaces the heuristic in the pipeline. And the labelled data, not the model, is the bottleneck: 73 percent of EduQG's labels are the lowest level, so a classifier that must recognise higher-order questions needs either more labelled examples of them or a coarser, better-populated label scheme. For the guide's quality-control purpose the practical reading is that Bloom's tags on remember and understand questions can be trusted, and tags on the higher levels should be treated as suggestions until the label supply improves. Both the model and the full metrics are saved (models/bloom_classifier/, outputs/bloom_classifier.json).

7.6 The trained claim extractor

The prompted claim extraction above was the declared stand-in for component 3, and the trained version now exists. I downloaded the Persuasive Essays 2.0 corpus from its official repository (402 essays, BRAT annotations for major claims, claims, and premises, with the corpus's own 322/80 train/test split) and fine-tuned DeBERTa-v3-base as a BIOES-style token classifier over the three component types. Sequences are paragraphs, since no annotated component crosses a paragraph and the longest paragraph fits inside 256 subwords, so nothing is truncated; evaluation is strict span-level (exact boundary and exact type) with sequence-level on the held-out test essays.

The result is a micro span-F1 of 0.63: premises 0.72, major claims 0.54, claims 0.44 (Figure 7.2). Strict matching is a hard yardstick, and the per-class order is the expected one, since claim boundaries are the classic ambiguity in this corpus. Dedicated argument-mining architectures with CRF decoding or joint relation modelling report higher figures, so this is a working first version rather than the state of the art, and I report it as such. For the pipeline it opens a design choice to settle with my supervisor: the trained extractor finds spans in the student's own words, while the prompted extractor produces readable claim paraphrases with sentence citations, so the natural combination is spans for provenance and the prompt for phrasing. The model and metrics are saved (models/claim_extractor/, outputs/claim_extractor.json).

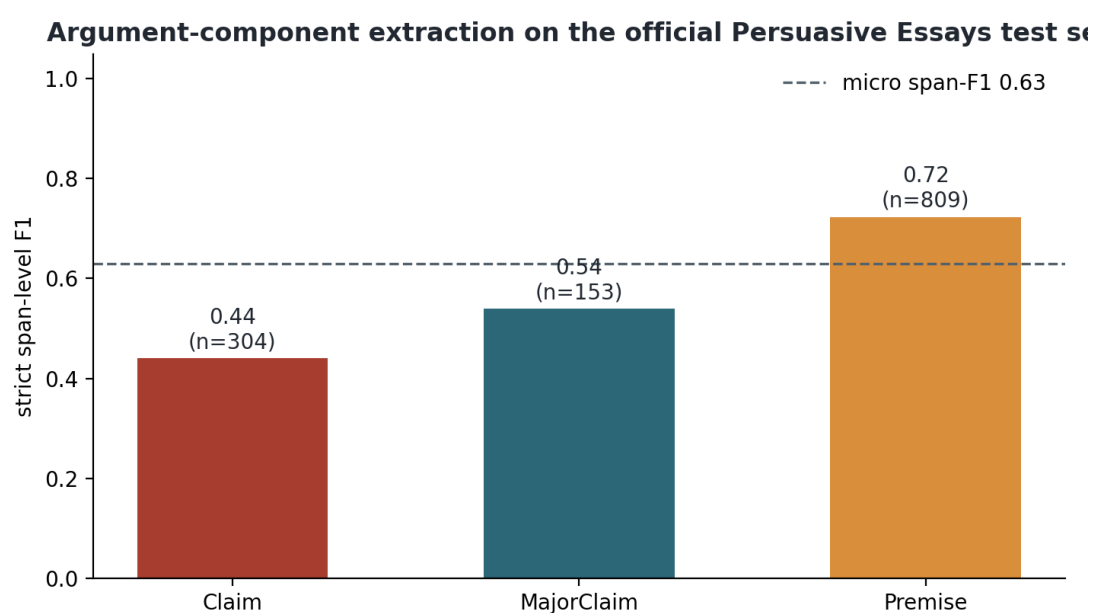


Figure 7.2: Argument-component extraction on the official Persuasive Essays test set, strict span-level F1 per class. Premises are learned well; claim boundaries are the hard case, matching the corpus literature.

7.7 The assembled guide: the pipeline's output document

With the classifier in place, the output assembler (`src/pipeline/assemble_guide.py`) now produces the document the pipeline exists for: a Verification Interview Guide a lecturer can take into a conversation. For a submission it runs the trained detector live and reports the probability with its limits stated in plain language (the in-domain F1, the out-of-domain false-positive risk, and the instruction to treat the score as a reason to talk, never as proof); it lists the SHAP-validated drivers of decisions in lecturer-readable terms; it presents each extracted claim quoted to its exact source sentences; it re-tags every question with the trained Bloom classifier, marking levels the classifier is weak on as advisory; and it closes with a suggested three-level rubric for reading the student's answers. The guide renders to Markdown, Word, and PDF (`outputs/verification_guides/3108a_ai_guide.pdf` is the worked example, generated end to end with live models). One design note carried through the document: the framing on the first page states that the guide is evidence for a conversation, not an accusation, which is the position the fairness results of Chapter 6 make necessary.

7.8 Fine-tuning the local backend: a result that looked too good

Section 7.4 left one question open for my supervisor: whether Backend B, the open-source side of the core comparison, should be the full Llama 3 8B with QLoRA or a smaller model that fits the laptop. The fit probes settled the practical half of it. In 4-bit with a LoRA adapter the 8B loads at 5.3 GB but only trains by spilling past the physical 8 GB into system memory, at 214 seconds per step for a 1024-token sequence, whereas Qwen2.5 3B trains inside VRAM at 2.8 seconds per step with about 3 GB of headroom (`outputs/qlora_fit_probe.json`). On the strength of those probes my supervisor signed off on Qwen2.5 3B as Backend B on 3 July 2026, so the fine-tuning experiment runs on the 3B.

I fine-tuned Qwen2.5-3B-Instruct (Yang et al., 2024) with QLoRA (Dettmers et al., 2023; `src/question_gen/finetune_qg.py`): 4-bit NF4 base, a LoRA adapter (rank 16; Hu et al., 2022) on the attention projections, gradient checkpointing and a paged 8-bit optimiser, batch 1 with gradient accumulation of 8, two epochs over 2,600 passage-to-question pairs from EduQG. The prompt tokens are masked in the loss, so the model learns only to produce the

question. Training finished in under an hour on the 4060 at a final loss of 1.42, and the adapter is saved to `models/qg_finetune_qwen3b/`.

The evaluation is built to isolate the fine-tuning effect and nothing else (`src/question_gen/eval_finetune.py`). One fixed set of 18 claims was extracted once from six flagged essays. The base Qwen 3B and the fine-tuned Qwen 3B each wrote verification questions for those same claims, and both sets were scored by the same discrimination simulation used in Chapter 8. The only thing that changes between the two conditions is the question-writing model, so any difference is the adapter. The three phases never share the 8 GB: claims are extracted with the local model through Ollama, which is then unloaded, each transformer model generates in turn, and scoring loads the embedding model last.

On the raw metric the fine-tune looks like a triumph. The base model's questions score a mean discrimination of 0.033 (95 percent CI 0.015 to 0.051); the fine-tuned model's score 0.154 (95 percent CI 0.085 to 0.227), a Mann-Whitney p of 0.007 and confidence intervals that do not overlap (Figure 7.3). Taken at face value that is roughly a five-fold gain, and the same 0.15 shows up again at essay scale in Section 8.8. I nearly wrote it up as the headline result of the whole project.

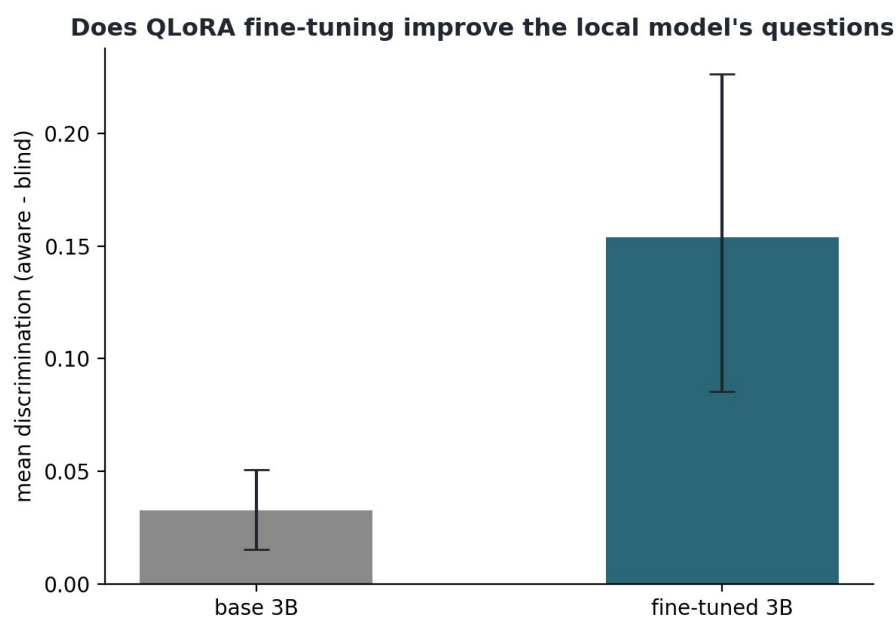


Figure 7.3: The QLoRA fine-tune of Qwen2.5 3B against its own base model on the same 18 claims, mean discrimination with 95 percent bootstrap intervals. The raw score rises from 0.033 to 0.154, but the quality audit in Section 8.9 shows this rise to be an artifact of degenerate output rather than better questions.

Reading the actual questions stopped that. The fine-tuned model had overfit the format of its training data. EduQG is largely a multiple-choice corpus, and the adapter learned to emit multiple-choice stems: about 95 percent of its questions are strings like "Which of the following is correct?" or "Which of the following is not a reason why Peugeot is successful?", with no options ever supplied, and a handful are raw JSON fragments leaking from the prompt. These are not verification questions at all. A student could not answer them, and a lecturer could not use them. The full audit across all four models is in Section 8.9, where the base 3B, the Llama 8B and the commercial model each produce zero degenerate questions and the fine-tuned model produces almost only degenerate ones.

Worse, those empty stems are precisely what drove the score up. The literal string "Which of the following is correct?" scores a mean discrimination of 0.44, higher than any real question from any model. It does so because a contentless question defeats the simulation: the source-aware and the source-blind answerer both receive a stem with nothing to answer, their responses diverge more or less at random, and the aware-minus-blind gap the metric reports reflects that randomness, not any property a real student would need the essay to satisfy. So the 0.154 measures how empty the fine-tuned model's questions are, not how good they are. The honest conclusion is the opposite of the raw number: the QLoRA fine-tune on EduQG failed as a question generator for this task.

I keep this in the dissertation rather than deleting it because the failure is informative twice over. First, it shows that the training data's format matters more than its size: two epochs on an overwhelmingly multiple-choice corpus was enough to collapse a capable instruct model into producing multiple-choice stems, so the fix is not more data but the right data, a question-generation set of open-ended questions (or the verification-style prompts themselves), which is the clear next experiment for Backend B. Second, it exposes a real weakness in the discrimination simulation: the measure assumes well-formed questions and can be gamed by degenerate ones, so it needs a well-formedness guard (a simple filter, or the Bloom classifier, rejecting non-questions) before its scores are trusted. That caution is exactly why the evaluation plan never rested on a single automatic metric, and Section 8.9 turns the episode into that methodological point.

Chapter 8: Evaluating the questions (discrimination simulation)

8.1 The idea

A good verification question is one that a student who understands their own work can answer and a student who does not cannot. I evaluate that without humans or an LLM judge, using a discrimination simulation (`src/evaluation/discrimination_sim.py`). For each question, one model answers it WITH the source passage from the essay (the "context-aware" answer) and another answers it with the question ONLY (the "context-blind" answer). I then measure how close each answer is to the source, using embedding similarity (nomic-embed-text; Nussbaum et al., 2024), and take the gap, aware minus blind, as the discrimination score. A high gap means the question genuinely needs the source; a near-zero gap means it can be answered without it. Everything runs on the local model and local embeddings, so the evaluation needs no API key and no judge.

8.2 The result, which surprised me

I expected my claim-grounded questions to discriminate well. They did not. On the flagged essay, the claim-grounded questions scored a mean discrimination of about 0.05 (95% confidence interval [0.01, 0.10]), while a set of generic essay questions ("what is your main argument", "what evidence most supports your conclusion", "explain your reasoning in your own words") scored about 0.31 ([0.27, 0.36]). The intervals do not overlap, so the difference is real, and it is the opposite of what I assumed (Figure 8.1).

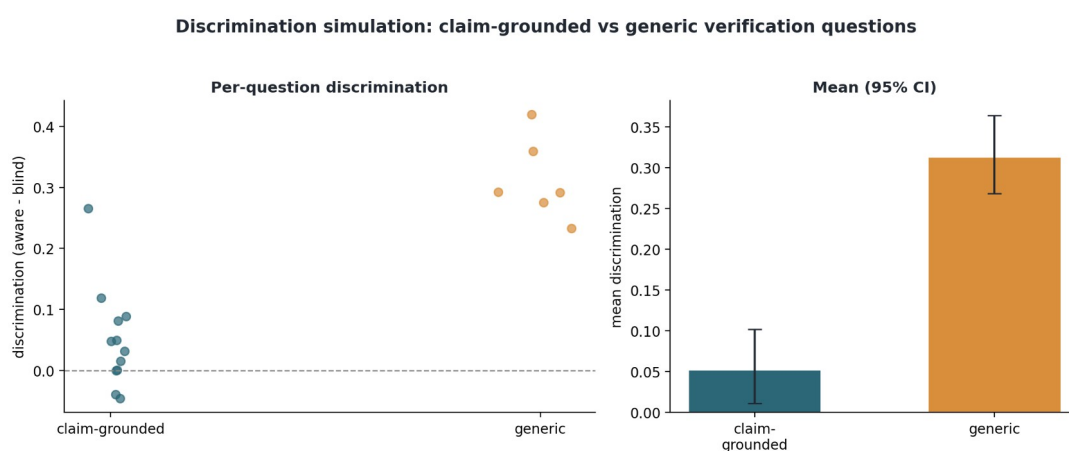


Figure 8.1: Discrimination simulation. Claim-grounded questions (left) cluster low; generic essay questions (right) discriminate much more. The gap is the source-aware minus source-blind answer similarity, with 95% confidence intervals.

8.3 Why, and what it means

The reason is the context-blind model. It is a large language model with broad world knowledge, so a question that names specific content, "what features make Shakespeare's dialogue intense", is answerable from general knowledge without ever reading this student's essay. The source barely changes the answer, so the discrimination is low. A generic question like "what is the main argument of your essay" is different: the blind model has no idea what this particular student argued, so its answer is vague, while the source-aware model answers about the actual essay. The source matters a lot, so the discrimination is high.

So the simulation is really measuring essay-specificity: does answering the question require this student's own essay, or only knowledge of the subject. That reframes what a good verification question is. It should force the student to reproduce their own argument, evidence, and choices, rather than recite facts about the topic that anyone knowledgeable could supply. When I rewrote the question-generation prompt to ask for the student's own reasoning and chosen evidence rather than general facts, the grounded score did improve, from about 0.03 to about 0.05, but it stayed well below the generic questions, because a specific question still tends to name the content and hand a knowledgeable answerer most of what it needs.

8.4 A caveat about the stand-in

The context-blind model is a strong stand-in, with far more world knowledge than a student who used AI without understanding the work. So a low discrimination score does not mean a question is useless in practice; it means the question is answerable by a knowledgeable person who never read this essay. The simulation is therefore a conservative, hard test of essay-specificity. A real student who did not understand their submission would do worse than the blind model on the grounded questions, so the true picture sits somewhere between the two lines in Figure 8.1.

8.5 What this changes

This is the first measured result for the question generation, and it is more useful as a finding than a score. It says the strongest verification questions are the ones that make the student retrieve and justify their own essay, its argument, the specific evidence they chose, and how their points connect, rather than questions that restate subject facts. The next iteration of the

generator will blend the claim grounding (so each question is tied to a real passage) with this retrieval-forcing style (so the answer cannot come from general knowledge). The remaining evaluation step is the supplementary LLM-as-judge rubric with cross-model agreement as a second view; the two steps promised here earlier, scaling the simulation across many essays and bringing in the commercial backend, are reported next.

8.6 Commercial versus local: the first scaled comparison

With the simulation in place, I ran the comparison at the centre of the research question. ATU could not provide API access or credits, so the commercial backend uses a free-tier Google Gemini API, plugged in behind the same interface as the local model; the local backend is the same Llama 3.1 8B used throughout. One disclosure on the commercial arm: free-tier quota limits forced a switch of Gemini endpoint partway through (four essays ran on `gemini-2.5-flash`, ten on `gemini-flash-latest`), so the commercial arm is one provider but not one fixed model snapshot. For each essay in a seeded random sample, each backend generated a Verification Interview Guide (three claims, grounded questions with sentence-level provenance), and every question from both backends was scored by the same local discrimination simulation, so what varies is which model wrote the questions.

A small pilot on the single flagged essay from Section 8.2 came first: there, both backends' grounded questions had discrimination intervals that included zero (local 0.032 with $[-0.005, 0.083]$; commercial 0.022 with $[-0.002, 0.048]$), too little signal to separate anything. The scaled run below is the real comparison.

Fifteen essays were sampled; fourteen produced a scored guide from both backends (one essay was excluded because claim extraction returned an empty guide; one commercial run was rate-limited for a day and completed on retry), giving each backend 126 questions. The pooled means are local 0.042 with a 95% interval of $[0.030, 0.055]$ against commercial 0.078 with $[0.058, 0.098]$. Because the same essays sit under both backends, the fair test is paired: the commercial backend scores higher on 10 of the 14 essays, with a mean paired difference of 0.036, bootstrap interval $[0.008, 0.067]$, paired t-test $p = 0.040$, and Wilcoxon signed-rank $p = 0.030$. Unlike in the pilot, both pooled means sit clearly above zero, so at this scale the grounded questions do measurably need the source (Figure 8.2).

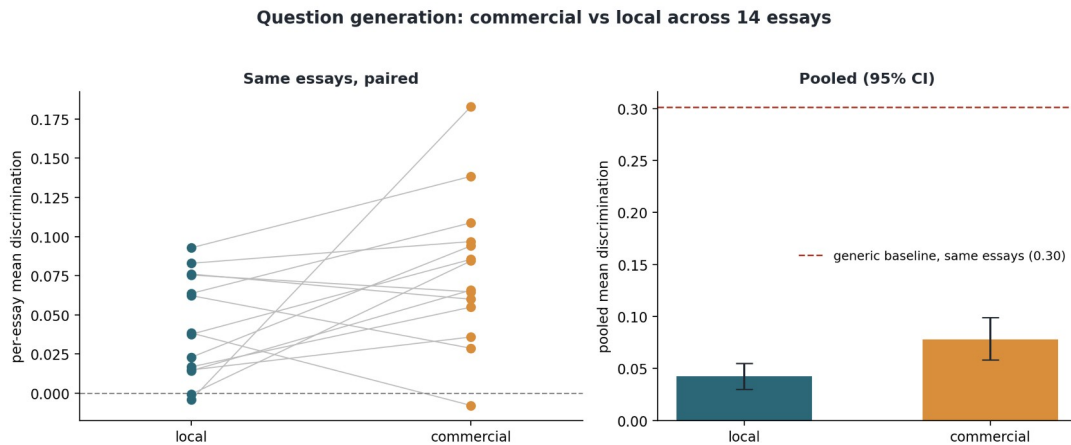


Figure 8.2: Commercial versus local question generation, balanced across the same 14 essays. Left: per-essay mean discrimination, paired. Right: pooled means with bootstrap 95% confidence intervals; the dashed line is the generic-question baseline.

The plain reading is that the commercial backend holds a small but statistically significant advantage on this measure, and that the local model remains competitive in absolute terms: the gap is about 0.04 against a generic-question baseline of 0.30 with a 95% interval of [0.28, 0.32], now re-measured on the same fourteen essays rather than taken from the pilot (84 generic questions; the pilot's single-essay value of 0.31 held up almost exactly), and it comes from a base model that has not yet been fine-tuned. Whether the planned QLoRA step closes that gap is exactly what the fine-tuning experiment will test. The sample-size trajectory is itself worth recording: nine essays showed no detectable difference, thirteen sat on the boundary ($t p = 0.053$), and the full fourteen crossed it. Small-n snapshots of this comparison would have supported whichever story one preferred, which is why the dissertation reports the complete set with paired tests. The remaining caveats stand: the discrimination measure is conservative, both backends sit well below the generic-question baseline, and the commercial arm is one provider's free tier rather than a frontier model. Extending the comparison to more essays, more providers, and the fine-tuned local backend is mechanical from here, since the framework saves per essay and resumes.

8.7 LLM-as-judge: three judges, anchored, and the anchoring was needed

The supplementary evaluation puts LLM judges over the questions with the four-dimension rubric (relevance, specificity, discrimination potential, cognitive appropriateness, each 1 to 5), and the plan has always been to validate judges rather than trust them. All three judges the scope calls for have now rated the twelve grounded questions from the pilot guide: Gemini (2.5 Flash, free tier), Claude (Opus 4.8) and GPT (4o-mini), the last two on a small

capped spend approved by my supervisor (`src/evaluation/llm_judge.py`, resumable across quota interruptions; results and the agreement statistics in `outputs/llm_judge.json`).

The three judges do not tell one story, and that is the finding. Gemini and GPT both sit at the ceiling: Gemini rates everything 4.5 to 5.0 (mean 4.81) and GPT even tighter, 4.75 to 5.0 (mean 4.94), so their scores mostly say the questions look well-formed. Claude behaves differently, using the scale from 2.5 to 4.5 (mean 3.67) and marking down exactly the content-naming questions such as "how does the metaphor evoke longing", which general knowledge can answer. Cross-model agreement, the scope's own validation criterion, is therefore poor: Krippendorff's alpha across the three judges is -0.25, and no pairwise rank correlation reaches significance (Spearman 0.20 to 0.45, all $p > 0.13$). Two judges at a ceiling cannot agree with a third that discriminates.

The anchoring to the objective measure settles what the disagreement leaves open. None of the three judges correlates positively with the discrimination simulation on the same questions: Gemini's mean rating sits at $\rho = -0.14$ ($p = 0.67$), Claude's at $\rho = -0.30$ ($p = 0.34$), and GPT's near-constant ratings manage a significantly negative ρ of -0.75 ($p = 0.005$), meaning the few questions it rated below the ceiling were among the ones that objectively discriminate best. With $n = 12$ the individual correlations deserve caution (an earlier partial run of the first judge looked strongly negative at seven questions and attenuated to nothing by twelve), but the pattern across three independent providers is consistent: rubric ratings measure how good a question looks, not how well it separates someone who read the essay from someone who did not.

The conclusion for the evaluation design is the one the plan anticipated, now with evidence rather than caution behind it. Judge scores in this setup cannot certify question quality on their own, because the judges neither agree with each other nor track the objective measure; the judge-free discrimination simulation carries the empirical weight, and the judge panel's value is exactly this negative result. It is also a second, independent instance of the project's central methodological lesson from Section 8.9: a plausible-looking automatic score, whether a judge rating or a discrimination number, means nothing until it is checked against something that cannot be gamed.

8.8 Like-for-like: the four writers on one fixed claim set

Section 8.6 compared the backends but let each one run its whole pipeline, so the model that chose the claims differed between arms as well as the model that wrote the questions. That leaves an ambiguity: the commercial edge there could come from writing better questions, or simply from Gemini selecting claims that happen to discriminate more. This section removes the ambiguity. One claim set is fixed per essay, extracted once with the neutral local extractor (Llama 3.1 8B), and four question writers answer the same claims: Llama 3.1 8B, free-tier Gemini, the base Qwen2.5 3B, and the QLoRA-fine-tuned Qwen2.5 3B. Every question is scored by the same discrimination simulation, so the only variable across the four arms is the model that writes the question (`src/evaluation/likeforlike_4way.py`, `outputs/likeforlike_4way.json`). The 14 essays are the balanced set from Section 8.6.

On the discrimination score the ordering looks dramatic (Figure 8.3). The fine-tuned 3B reaches a pooled mean of 0.153 (95% CI [0.105, 0.202]), far above the base 3B at 0.041 [0.028, 0.055], Llama 8B at 0.031 [0.016, 0.045], and commercial Gemini at 0.024 [0.008, 0.041]. Its paired lead is 0.138 over commercial ($p < 0.001$) and 0.123 over its own base ($p < 0.001$), and it reproduces the 0.154 from Section 7.8. Read at face value, the locally fine-tuned open model does not merely match the commercial one, it buries it.

That reading does not survive looking at the questions, and Section 8.9 is given over to why: about 95 percent of the fine-tuned model's questions are degenerate multiple-choice stems that game the simulation, so the 0.153 measures how empty they are, not how good. The fine-tuned arm is therefore set aside for the interpretation here, and the honest comparison is among the three models whose questions are well-formed.

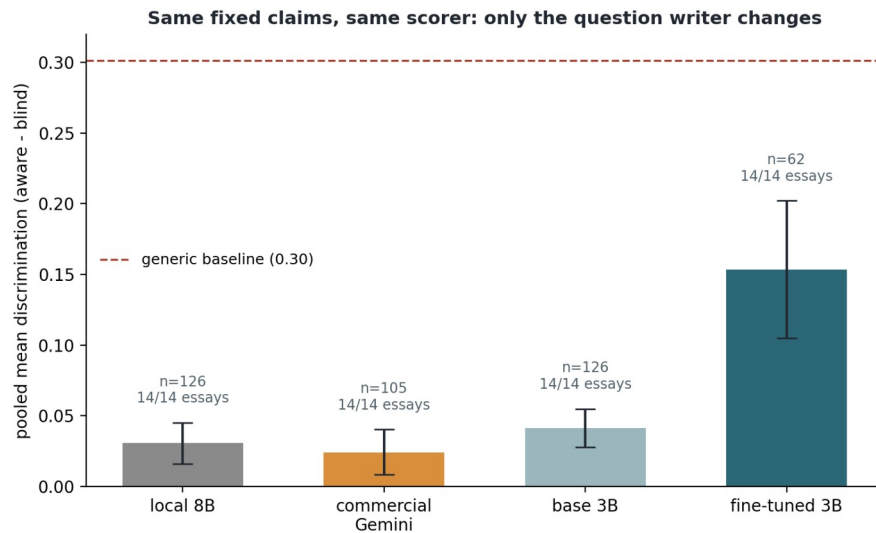


Figure 8.3: The four question writers on one fixed claim set per essay, same scorer, pooled mean discrimination with 95% bootstrap intervals, each bar labelled with its question count and essay coverage. The fine-tuned 3B posts the highest raw score, but Section 8.9 shows that score to be an artifact of degenerate questions; the meaningful comparison is among the other three arms.

Among those three the picture is modest and close. On the fixed claims the three models are statistically indistinguishable: base Qwen 3B at 0.041, Llama 8B at 0.031, and Gemini at 0.024, all low and with overlapping intervals. The run initially covered only 9 of 14 essays on the commercial arm because free-tier quota ran out mid-run; the framework resumed when quota returned, and the numbers here are the complete set (105 commercial questions across all 14 essays, a few claims lost to rate-limiting inside otherwise-covered essays). The complete set settles what the partial one could only suggest. In Section 8.6, with each backend choosing its own claims, Gemini held a significant edge over Llama 8B (0.078 to 0.042, paired $p = 0.040$); here, on identical claims, that edge is simply gone (paired difference -0.005, $p = 0.62$, Gemini higher on seven of fourteen essays, a coin flip). So the Section 8.6 advantage does not survive fixing the claims. The reading I take is that it came substantially from claim selection, Gemini choosing easier-to-discriminate claims when allowed to, rather than from writing stronger questions on the same ones; the design here cannot rule out every alternative, but the disappearance itself is now measured on the full sample, not hypothesised. The firm conclusion of this section is narrower than the one I first drafted and arguably more useful: when the task is fixed, small open models running on a laptop write questions as discriminative as a commercial model's, while the second half of the research question, whether a fine-tuned local model can beat a commercial one, stays open at this

point in the story, because the fine-tune that was supposed to answer it produced unusable questions.

8.9 The fine-tuned model games the metric: a quality audit

The previous section leaned on a number without reading what it scored. This section does the reading, because the gap between the two is the most important methodological lesson in the project (`src/evaluation/qg_quality_audit.py`, `outputs/qg_quality_audit.json`). Every question from all four arms was checked with a transparent rule for degeneracy: multiple-choice stems (which never carry options here), raw JSON fragments leaking from the prompt, and contentless "which is correct" forms.

The result is stark (Figure 8.4). The Llama 8B, base 3B and commercial arms produce no degenerate questions at all. The fine-tuned 3B produces almost nothing else: 59 of its 62 questions, just over 95 percent, are stems like "Which of the following is correct?" or "Which of the following is not a reason why Peugeot is successful?", with no options supplied. The fine-tune, two epochs on the largely multiple-choice EduQG corpus, had overfit the format of the training data and forgotten how to write an open question. None of these are usable in a verification interview.

The second panel is the part that matters for the evaluation method. Those empty stems do not merely survive the discrimination simulation, they win it. The single literal string "Which of the following is correct?", scored eleven times across the run, averages a discrimination of 0.44, higher than the mean of every model's real questions. The reason is structural: a contentless question gives the source-aware answerer and the source-blind answerer nothing to latch onto, their answers wander apart for reasons unrelated to the source, and the aware-minus-blind gap the metric rewards is large. The discrimination simulation, in other words, silently assumes the questions it scores are well-formed, and a degenerate generator can exploit that assumption to post an arbitrarily high score.

Why the fine-tuned model's discrimination score is an artifact

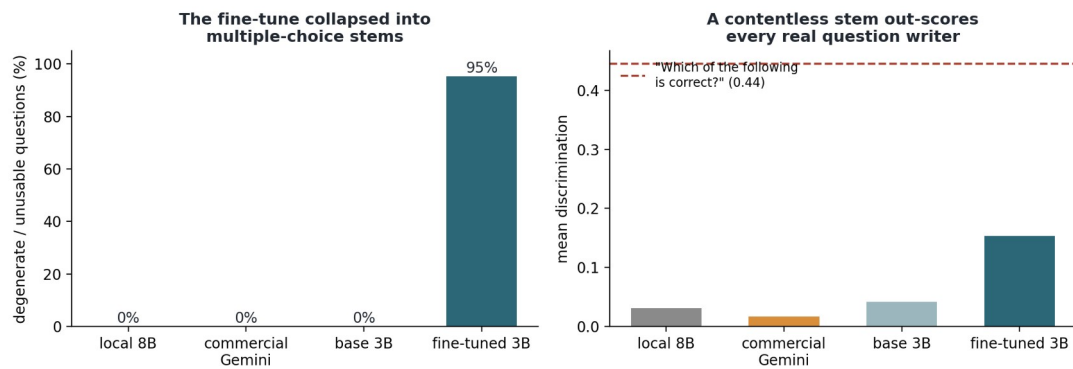


Figure 8.4: Auditing question quality behind the discrimination score. Left: the share of degenerate, unusable questions per model. Right: the mean discrimination of each model's questions against the score of the contentless stem "Which of the following is correct?" (dashed line), which out-scores every real question writer.

Two conclusions follow, and both are more useful than the false headline they replace. For Backend B, the lesson is that the training data's format dominates: fixing the fine-tune means not more data but the right data, an open-ended question-generation set (SQuAD-style questions, or the pipeline's own verification prompts distilled into training pairs) rather than a multiple-choice corpus, and that is the next experiment, carried out in Section 8.10. For the evaluation, the lesson is that the judge-free discrimination simulation, valuable as it is for well-formed questions, needs a well-formedness gate in front of it, and that no automatic score should be read without inspecting the text behind it. That gate is now part of the pipeline (`src/question_gen/wellformed.py`): the same transparent rule the audit uses is applied wherever questions leave the system, the guide builder drops malformed questions before a lecturer can see them and records how many it dropped, and every fine-tune evaluation reports its degeneracy rate next to its score. That is precisely the discipline the evaluation plan was built on, which is why this failure was caught here rather than in a viva.

8.10 Fixing the fine-tune: v2 on open-ended data

Section 8.9 made a testable prediction: if the degeneracy was caused by EduQG's multiple-choice format, then changing only the training data to open-ended questions should fix it. So I re-ran the fine-tune with everything held identical except the data, swapping EduQG for SQuAD (Rajpurkar et al., 2016), whose questions are open-ended and passage-grounded (`src/question_gen/finetune_qg_v2.py`, `adapter` in `models/qg_finetune_qwen3b_v2/`). The evaluation repeats the Section 7.8 design on the

same 18 claims, base against the new adapter, and it audits degeneracy before reading any score (src/question_gen/eval_qg_v2.py, outputs/qg_v2_eval.json).

The prediction holds, cleanly. The v2 model's questions are 0 percent degenerate against v1's 95, so the format of the training data was indeed the cause, not fine-tuning as such (Figure 8.5). And this time the discrimination gain is real rather than an artifact, because the questions are well-formed. On the same claims, v2 scores a mean discrimination of 0.102 (95% CI [0.060, 0.144]) against the base model's 0.027 ([0.015, 0.040]), a Mann-Whitney p of 0.0003. The contrast with v1 is the whole point: v1's higher 0.154 was empty multiple-choice stems gaming the metric, while v2's lower 0.102 is genuine questions that genuinely need the source more than the base model's do.

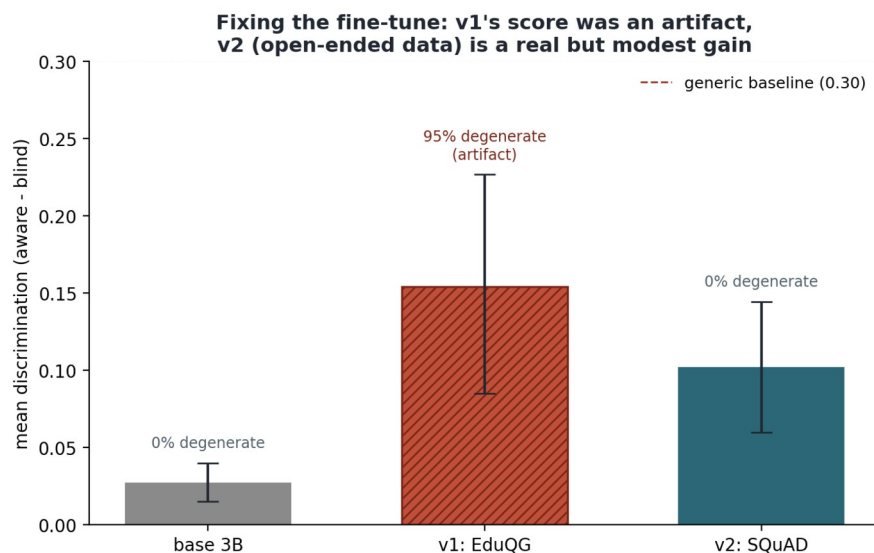


Figure 8.5: The fine-tune, corrected. Base 3B, the v1 EduQG adapter (hatched: its score is an artifact of 95 percent degenerate questions), and the v2 SQuAD adapter, mean discrimination on the same 18 claims with 95 percent intervals and each bar's degeneracy rate. v2 is a real gain over base ($p = 0.0003$) on well-formed questions, but modest and still below the generic-question baseline.

I hold the reading to what was measured, because v2 is a partial success, not a triumph. Its 0.102 is well under the generic-question baseline of about 0.30, and reading its questions shows why: SQuAD trains a factual style ("What is the purpose of judicial review?"), and factual questions are partly answerable from general knowledge, which is exactly the property Section 8.3 showed the simulation penalises. Two smaller points round out the honesty. v2 needed a cleaned output parser, because the adapter sometimes appended JSON fragments after the question that an earlier extractor would have kept; the parser now takes the first well-formed question and drops the rest. And the degeneracy audit ran before the

score, as the standing rule now requires. The overall arc is the useful result: fine-tuning a small local model does help once the training data has the right form, but SQuAD's factual questions are not the ideal signal for verification, so the genuinely next step is a training set of reasoning-demanding questions, distilled from the pipeline's own verification prompts, which is the v3 experiment of the next section. That is where the arc stood before its final leg: a diagnosed and half-fixed component, reported honestly rather than as the false win of Section 7.8's first draft.

8.11 Completing the data-format experiment: v3 on the pipeline's own style

The third leg trains on data in exactly the format the task wants. A training set of 2,604 verification questions was distilled from the pipeline itself (`src/question_gen/build_v3_dataset.py`): the local Llama 3.1 8B ran the production claim-extraction and question-writing prompts over 307 training essays, and each well-formed question became one passage-to-question pair, with the passage being the claim plus its source sentences, which is what the model sees at inference. Three precautions mattered. All fifteen evaluation essays were excluded from the training pool, so nothing the fine-tunes are tested on was ever trained on. Only questions passing the well-formedness gate became training targets, since a noisy teacher would poison the experiment. And Llama's licence permits training other models on its output, which is why the teacher is local rather than a commercial API. Training kept every setting identical to v1 and v2 once again (`src/question_gen/finetune_qg_v3.py`); the final loss of 0.72, against 1.42 for v1 and 1.27 for v2, already hints that on-format data is simply easier to learn.

The evaluation follows the standing protocol: degeneracy audited before any score, same 18 fixed claims (`src/question_gen/eval_qg_v3.py`, `outputs/qg_v3_eval.json`). v3 is fully well-formed, zero degenerate questions out of 42, and it answers the request for three questions per claim far better than v2 did (2.3 per claim against 1.3). Its output finally reads like the pipeline's design language: "How did you decide to use the example of a decision made by a public authority as evidence", "what role does the concept of authority play in your argument, and how did you connect it", "in what ways does your discussion relate to the broader themes of your essay". On the metric it lands at a mean discrimination of 0.064 (95% CI [0.033, 0.096]), roughly two and a half times the base model's 0.027 ([0.015, 0.040]), with the intervals barely brushing; notably that also doubles the 8B teacher's own score in the

fixed-claim comparison (0.031, Section 8.8), so the 3B student overtook the model that taught it (Figure 8.6).

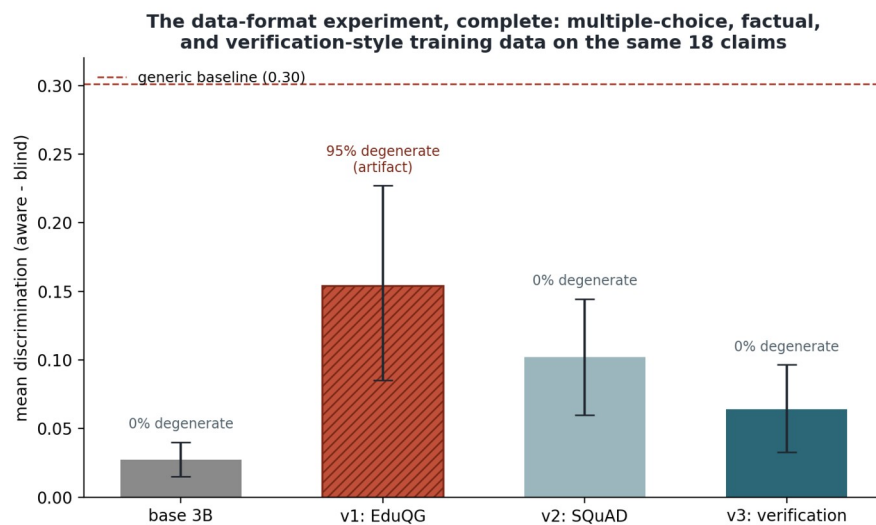


Figure 8.6: The data-format experiment complete on the same 18 claims: base model, v1 (multiple-choice EduQG, hatched because its score is a degeneracy artifact), v2 (factual SQuAD) and v3 (verification-style self-distillation), with each bar's degeneracy rate. Both open-ended formats produce fully well-formed output and real gains; only the multiple-choice format collapses.

The result that resists a tidy ranking is that v2 still posts the higher number (0.102 against 0.064), and saying why matters more than the numbers themselves. The two adapters learned different trades. v2 writes one terse factual question per claim; that style happens to suit the simulation, but "What is the purpose of judicial review?" is not a verification question, since it never asks the student about their own essay. v3 writes the questions the product actually needs, demanding the student's reasoning, choices and connections, but in doing so it quotes the claim's content back, and Section 8.3 established that handing content to the source-blind answerer shrinks the measured gap. So the metric, taken alone, would pick the adapter whose output is unfit for purpose. This is the third time in this chapter a single automatic number pointed the wrong way (after the v1 artifact and the judge panel), and it is why the decision about Backend B is not delegated to the metric: v3 becomes the working local backend on the grounds that its output is well-formed, on-style and a real improvement over base, with v2's higher raw score documented and explained rather than chased.

What the completed experiment establishes is worth stating cleanly, because it is the dissertation's answer on fine-tuning. Format dominates: a multiple-choice corpus collapses the model, and either open-ended format fixes it. Fine-tuning a 3B model on an 8 GB laptop reliably lifts it well clear of its base, whichever open format is used, and self-distillation can

lift a small student past its own larger teacher on the target measure. And no single automatic score, not this one and not a judge's, is allowed to settle a design decision by itself in this project.

Chapter 9: Discussion

9.1 Answering the research question

The research question had two halves: whether an explainable pipeline can support lecturers in academic integrity verification by combining transparent detection with argument-aware question generation, and how far locally fine-tuned open-source models can match commercial LLMs at it. The work gives each half a concrete answer, and neither answer is the simple one I expected at the start.

On the first half, the pipeline exists and works end to end. A submission goes in; a detector trained on a matched, audited corpus scores it; the score arrives with a faithfulness-tested explanation in lecturer language rather than a bare percentage; the submission's own claims are extracted with sentence-level provenance; verification questions grounded in those claims are generated, gated for well-formedness, tagged with a cognitive level, and assembled into an interview guide whose first page says what the whole design says: this is evidence for a conversation, not an accusation. Every stage was evaluated, and the evaluations are the reason the answer is credible. The detection is excellent in domain (F1 0.990 with a bootstrap interval of roughly [0.97, 1.00]) and transfers across unseen generators (0.97), but degrades across text domains (0.79) in the direction that matters most, falsely flagging human academic writing. The explanation layer is honest about itself: all three token-level methods, attention included, fail a faithfulness test that the feature-level SHAP account passes, so the feature account is the one a lecturer sees.

On the second half, the answer turned out to depend on holding the task fixed, which is itself a finding. When each backend chose its own claims, the commercial model held a small significant edge (paired difference 0.036, $p = 0.040$). When every model wrote questions for identical claims, the edge vanished (difference -0.005, $p = 0.62$): most of the apparent commercial advantage was claim selection, not question writing. Fine-tuning then moved the local backend well clear of its own base once the training data had the right format, and the fine-tuned 3B model doubled the score of the 8B model that taught it. Put together, the honest answer to "can locally fine-tuned open models match commercial LLMs at this task" is yes on this evidence, with the caveat that the margin between all well-formed question writers is small against the generic-question baseline, and the deeper point that the

comparison itself is fragile: its outcome flipped with the experimental design, which is why the dissertation reports both designs rather than the friendlier one.

9.2 What the failures taught: trust nothing you have not checked

The most valuable results in this dissertation are the ones that nearly deceived me, because the project's thesis is that unexplained scores cannot be trusted, and three times my own scores proved it.

The first was the fine-tune artifact. The v1 adapter posted a discrimination score four times its base model, with non-overlapping confidence intervals, and for about an hour it was the headline of this dissertation. Reading its actual output showed 95 percent degenerate multiple-choice stems, and the audit showed those stems do not merely survive the metric but win it: a contentless question makes the source-aware and source-blind answers diverge at random. The second was the judge panel. Three commercial judges, the validation the scope called for, produced ratings that neither agree with each other (Krippendorff's alpha of -0.25) nor track the objective measure, with one judge significantly anti-correlated. Rubric ratings measure how good a question looks. The third was quieter but the same shape: ranking the fine-tunes by raw discrimination would have picked v2, whose terse factual one-liners are not verification questions at all, over v3, whose output is exactly what the product needs.

One methodological posture answers all three, and it is now built into the pipeline rather than stated as advice. Every automatic score is anchored to something it cannot game: the judges are anchored to the simulation, the simulation is gated for well-formedness with the degeneracy rate reported next to every score, and no design decision was delegated to a single number. I would argue this posture, demonstrated three times on the project's own results, is as much a contribution as any component, because it is precisely the discipline the field's detector vendors do not practise.

9.3 Verification over accusation

The fairness evidence shaped the design more than any other single input. The literature reports that detectors flag non-native English writers at wildly disproportionate rates, and my own robustness tests reproduced the pattern's mechanism at domain level: human arXiv abstracts were falsely flagged 79 percent of the time because dense academic prose looks, to

these features, like machine text. A detector with that failure mode cannot be the end of any fair process. The pipeline therefore treats detection as the opening of a conversation and moves the evidential weight to verification: questions a student who wrote and understood the work can answer and an impostor cannot, each question traceable to exact sentences of the submission so nothing is invented.

The discrimination simulation sharpened what such questions must look like. Its central finding is that content-naming questions are answerable from general knowledge, so good verification questions force the student to reconstruct their own reasoning, name their own evidence, and connect their own argument. That principle survived every subsequent experiment, reshaped the generation prompt, drove the v3 training data, and explains the ranking twist between v2 and v3. It also bounds what the simulation can see: the context-blind model is a much stronger stand-in than a student who submitted work they do not understand, so measured discrimination is a conservative floor, not an estimate of classroom performance.

9.4 Limitations

The limitations are stated throughout the chapters; gathering the important ones in one place keeps the claims honest. The evaluation of question quality rests on a simulation, not on students; that was a deliberate scope decision (it keeps the project clear of human-participants approval), but it means the discrimination scores are proxies with a conservative bias, and classroom validation is the single most important piece of future work. Sample sizes are modest: fourteen essays and eighteen fixed claims in the comparisons, twelve questions in the judge study; paired designs and bootstrap intervals are used throughout, and the sample-size trajectory of the main comparison is reported precisely because small snapshots would have supported whichever story I preferred. The detection corpus pairs one human corpus with one generator, and although the detector transfers to six unseen generators, the corpus itself cannot show generator diversity. The commercial arm is one provider's free tier rather than a frontier model, and quota limits caused disclosed irregularities. On the home corpus the detection task saturates, which is why the training-distribution control could show the balancing is harmless but not whether it helps on a harder task. And the argument-mining component extracts claims at a working but unspectacular span-F1 of 0.63, and relation

classification between claims and premises is not built; whether to add a basic version or record a descope is an open question with my supervisor at the time of writing.

9.5 Implications for practice

For a lecturer, the practical output of this work is a defensible process rather than a score. A flag arrives with named, tested reasons ("the sentences are unusually uniform, the vocabulary less varied than a typical student's"), and with an interview guide rather than a verdict. For an institution, two results matter. First, detectors, including good ones, must not be pointed at text from domains they were not trained on, and any deployment needs domain checks, calibrated thresholds and an abstain band, because the failure mode is false accusation and it lands unevenly. Second, the local-versus-commercial result removes a dependency: models that run on a single consumer laptop matched a commercial API on the fixed task, which means an institution can run verification without sending student work to a third party, a real consideration given that submissions are personal data. The pipeline was built entirely on that constraint, and the constraint turned out to be a feature.

Chapter 10: Conclusions and future work

10.1 What this dissertation contributes

The project set out to replace an indefensible number with a defensible process, and to find out whether that process needs a commercial API behind it. Its contributions, each backed by a result in the preceding chapters, are these.

First, a matched detection corpus and an audited detector. The corpus pairs 640 real student essays with 640 AI essays matched on topic and length, so the classes differ in writing rather than in anything a detector could cheat on; the audit methodology that caught a corpus artefact behind an initially perfect score, removed it from both classes, and re-established the result on style alone is reusable beyond this project.

Second, an explanation stack whose explanations are tested, not assumed. Three attribution methods were measured on one ablation yardstick; the two token-level methods fail it for the same reason, a diffuse signal, and the feature-level SHAP account passes it and becomes the lecturer-facing explanation. The claim "this explanation is faithful" is, throughout, an empirical statement.

Third, an argument-aware question-generation pipeline with provenance by construction: claims cite sentence numbers that are looked up rather than quoted, so a generated question can always be traced to exact lines of the submission, and a well-formedness gate keeps degenerate output away from both the lecturer and the metrics.

Fourth, a judge-free evaluation for verification questions, and the demonstration that it can be gamed. The discrimination simulation measures whether a question needs the source; the discovery that contentless questions defeat it, and the gate that closes that hole, are a methods contribution to an evaluation literature that leans heavily on LLM judges.

Fifth, the three-judge validation with a decisive negative result: commercial judges neither agree with each other nor track the objective measure on this task, which is direct evidence for anchoring judge studies rather than trusting them.

Sixth, the commercial-versus-local comparison run under two designs, showing the comparison's outcome is a property of the design: a significant commercial edge under self-

chosen claims disappears entirely under fixed claims. On the fixed task, models running on one consumer laptop matched the commercial API.

Seventh, the data-format experiment across three fine-tunes of the same model with identical settings: a multiple-choice corpus collapses the model into unusable output, either open-ended format repairs it, the verification-style data produces the on-style backend, and the 3B student overtook its 8B teacher on the target measure.

And eighth, the working artifact itself: a Verification Interview Guide that generates end to end from a submission with live models on an 8 GB laptop, framing its contents as evidence for a conversation.

10.2 Future work

The clearest next step is the one this project deliberately excluded: a study with real students, under ethics approval, testing whether the generated questions separate authors from non-authors in practice as the simulation predicts. The simulation's conservative bias makes this a hopeful experiment rather than a hail-Mary, and the guide is the ready-made instrument for it.

Several extensions follow directly from measured limits. The commercial arm should widen to multiple providers and a frontier model now that the framework treats any API as a plug-in backend. The detection corpus should gain generations from more model families, letting the transfer results rest on the corpus itself rather than an external benchmark. Cross-domain deployment needs calibrated per-domain thresholds and an abstain band before anyone should trust a flag outside student essays, and the hybrid detector's out-of-domain behaviour suggests where those calibrations should start. The training-distribution control should be re-run on a task hard enough to have resolution. The Bloom classifier is throttled by label supply, not architecture, so a better-populated labelling of higher-order questions would lift the guide's quality control. Relation classification between claims and premises would complete the argument-mining component as originally specified. And the agreed future extension of the classification target, a third partial-AI class, becomes worth piloting once mixed human-and-AI documents can be constructed carefully, since partial assistance is the realistic case in coursework.

10.3 Closing

The dissertation's method can be stated in one sentence: never present a number you cannot explain, and never trust one you have not tried to break. Practising that on my own results cost me three headlines, an artifact fine-tune, an agreeing panel of judges, and a tidy ranking, and each time it replaced a false claim with a finding that was more useful and easier to defend. That, more than any single score, is what I would want a lecturer, or an examiner, to take from this work: the pipeline is trustworthy in exactly the places where it can show its working, and it is honest about the places where it cannot.

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